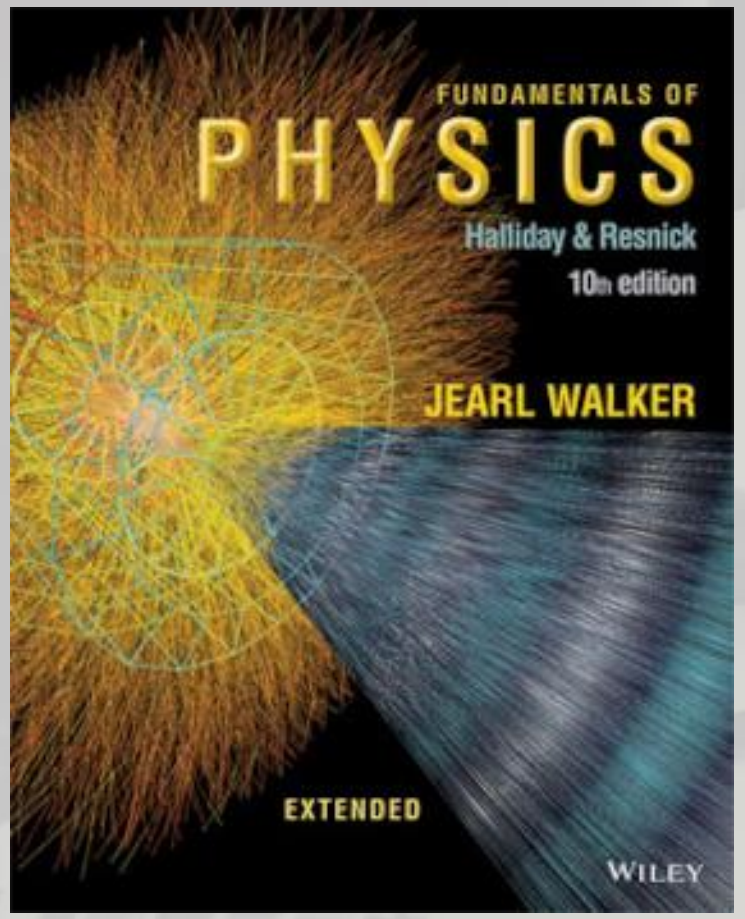
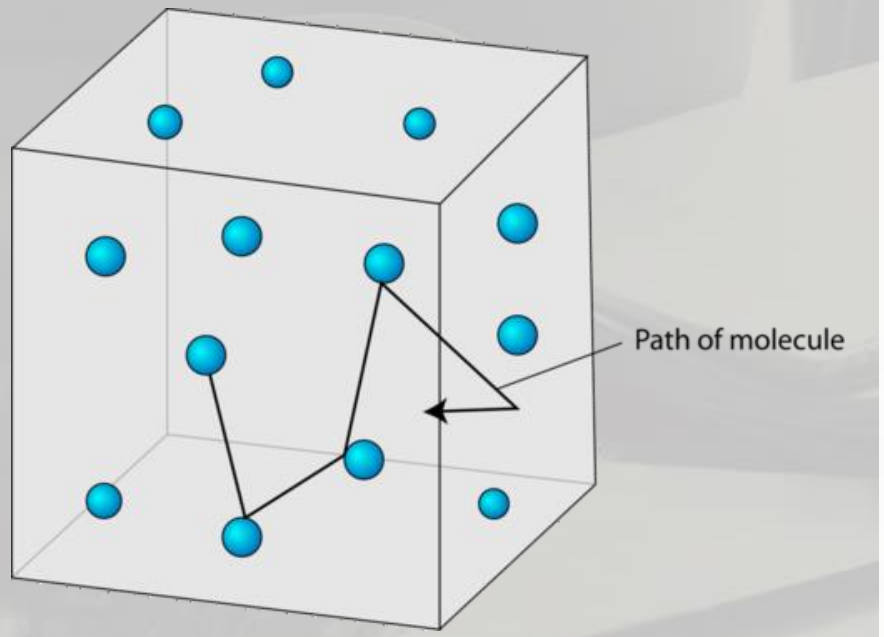
III. Kinetic theory of gases



Outcomes

- Apply the ideal gas law
- Apply relationship between pressure, temperature, kinetic energy and speeds of molecules
- Apply relationship between mean free path, diameter of molecules, and number of molecules per unit volume
- Apply the molar specific heats and of adiabatic expansion an ideal gas

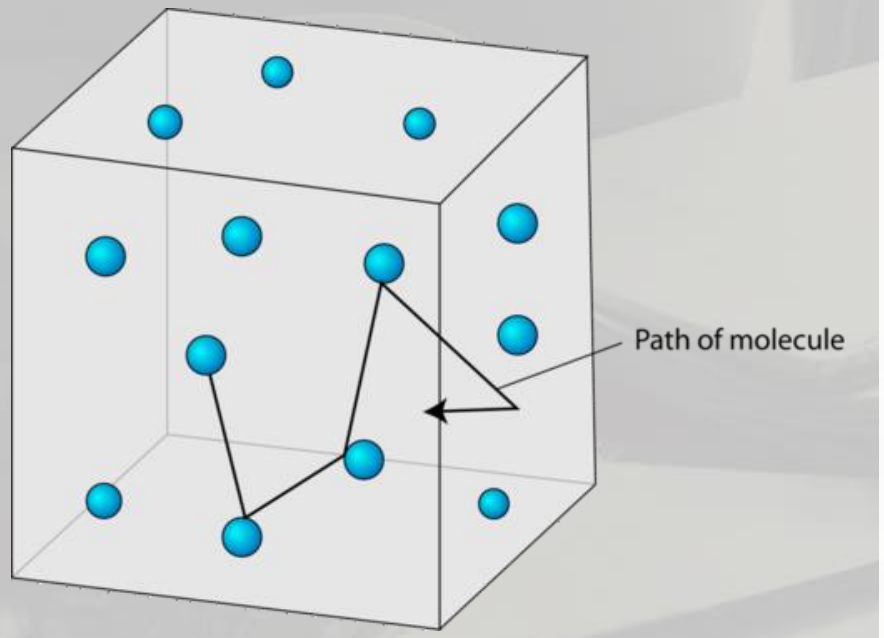
III. Kinetic theory of gases



Introduction

- One of the main subjects in thermodynamics: **physics of gases**
- Three variables associated with a V, T, P—are all a consequence of the motion of the atoms
- Kinetic theory of gases: A model of behavior of gases that relates the motion of the atoms to V, T, P of the gas

III. Kinetic theory of gases

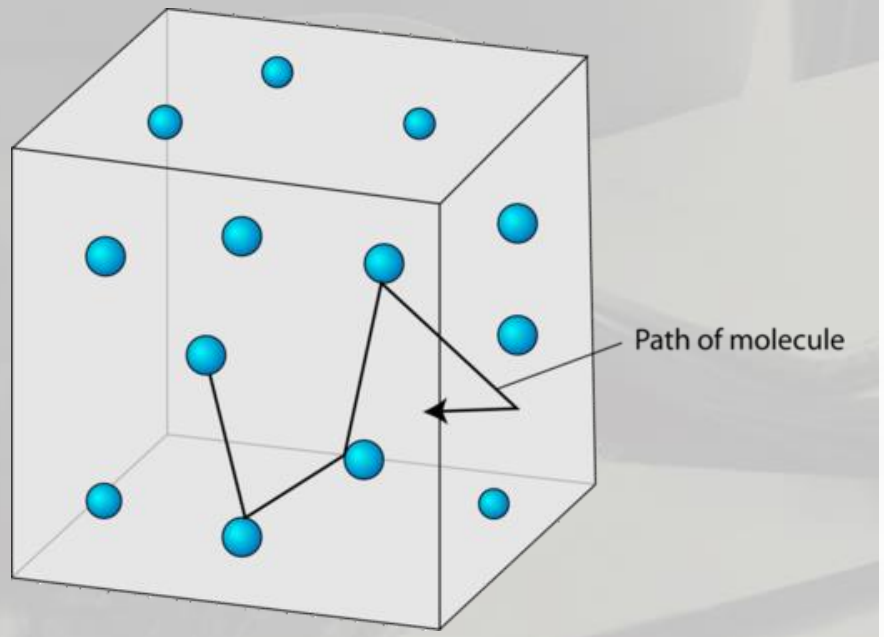


Introduction

Assumptions:

- The gas consists of very small particles
- The particles have the same mass
- The number of particles is so large that statistical treatment can be applied
- All collisions are perfectly elastic
- Except during collisions, the interactions among molecules are negligible.

III. Kinetic theory of gases

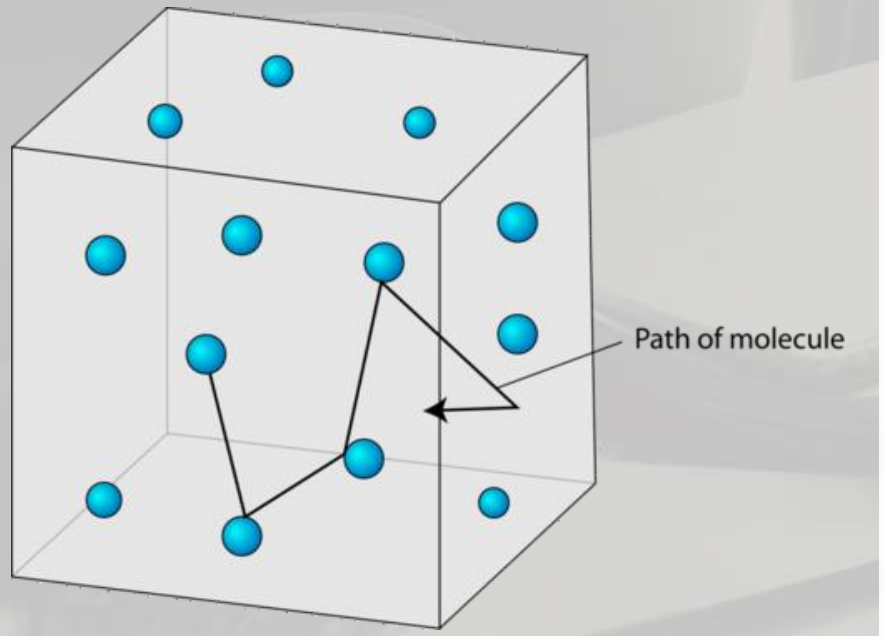


Introduction

Some applications:

- The combustion of vaporized fuel (a gas) in the automobile engines
- The production rate of the fermentation gas that causes bread to rise as it bakes
- How heat exchanges between the oceans and the atmosphere can affect weather conditions
- Etc.

III. Kinetic theory of gases

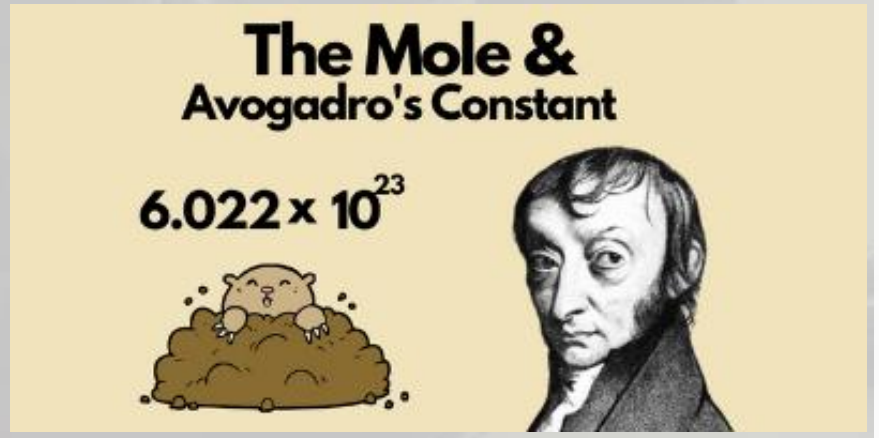


Introduction

Four main relationships:

- Pressure and kinetic energy
- Temperature and kinetic energy
- Collision with container
- Speeds of molecules

Avogadro's number



One mole?

- « Old definition»: The number of atoms in a 12 g sample of carbon-12
- «New definition»: One mole contains exactly 6.022 140 76 × 10^{23} elementary entities

$$N_A = 6.02 \times 10^{23}\ mol^{-1}$$

- Amedeo Avogadro (1776–1856): "*all gases occupying the same volume under the same conditions of temperature and pressure contain the same number of atoms or molecules*"

Avogadro's number

$$n = \frac{N}{N_A}$$

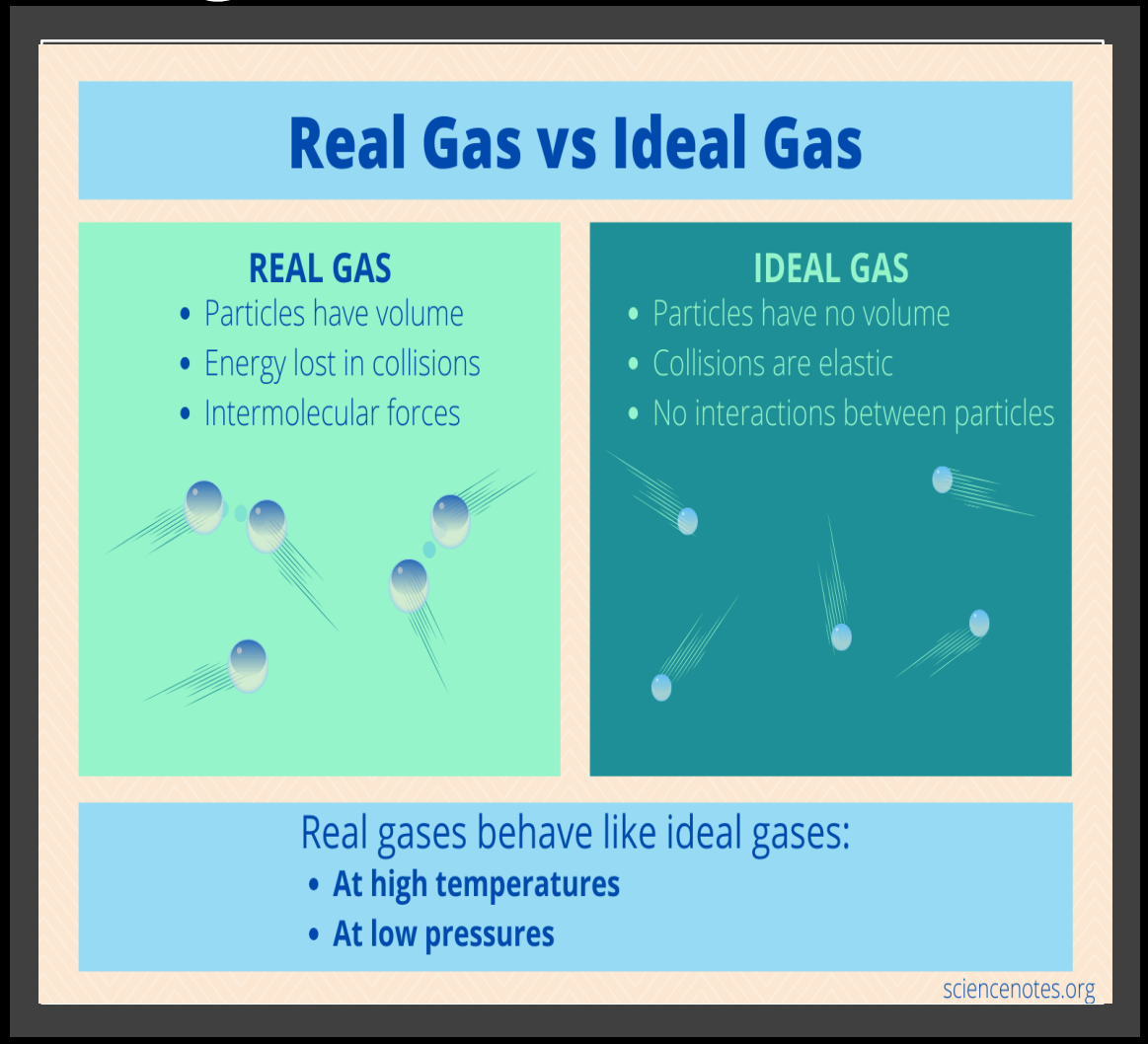
n: the number of moles
N: the number of molecules
N_A: the number of molecules in 1 mol

$$n = \frac{M_{sam}}{M} = \frac{M_{sam}}{mN_A}$$

M_{sam}: the mass of the sample
M: the *molar mass* (the mass of 1 mol)
m: the molecular mass

The number of moles n contained in a sample of any substance is equal to the ratio of the number of molecules N in the sample to the number of molecules N_A in 1 mol

Ideal gas law



- Goal: explain macroscopic properties of a gas in terms of behavior of the molecules that make it up
- If we confine 1 mol samples of various gases in boxes of identical V and T, their measured P are almost the same, and **at lower densities the differences tend to disappear**.
- All real gases approach the ideal state at **low enough densities**
- *Ideal gas* concept allows to gain insights into the limiting behavior of real gases.

Ideal gas law (general gas equation)

$$PV = nRT$$

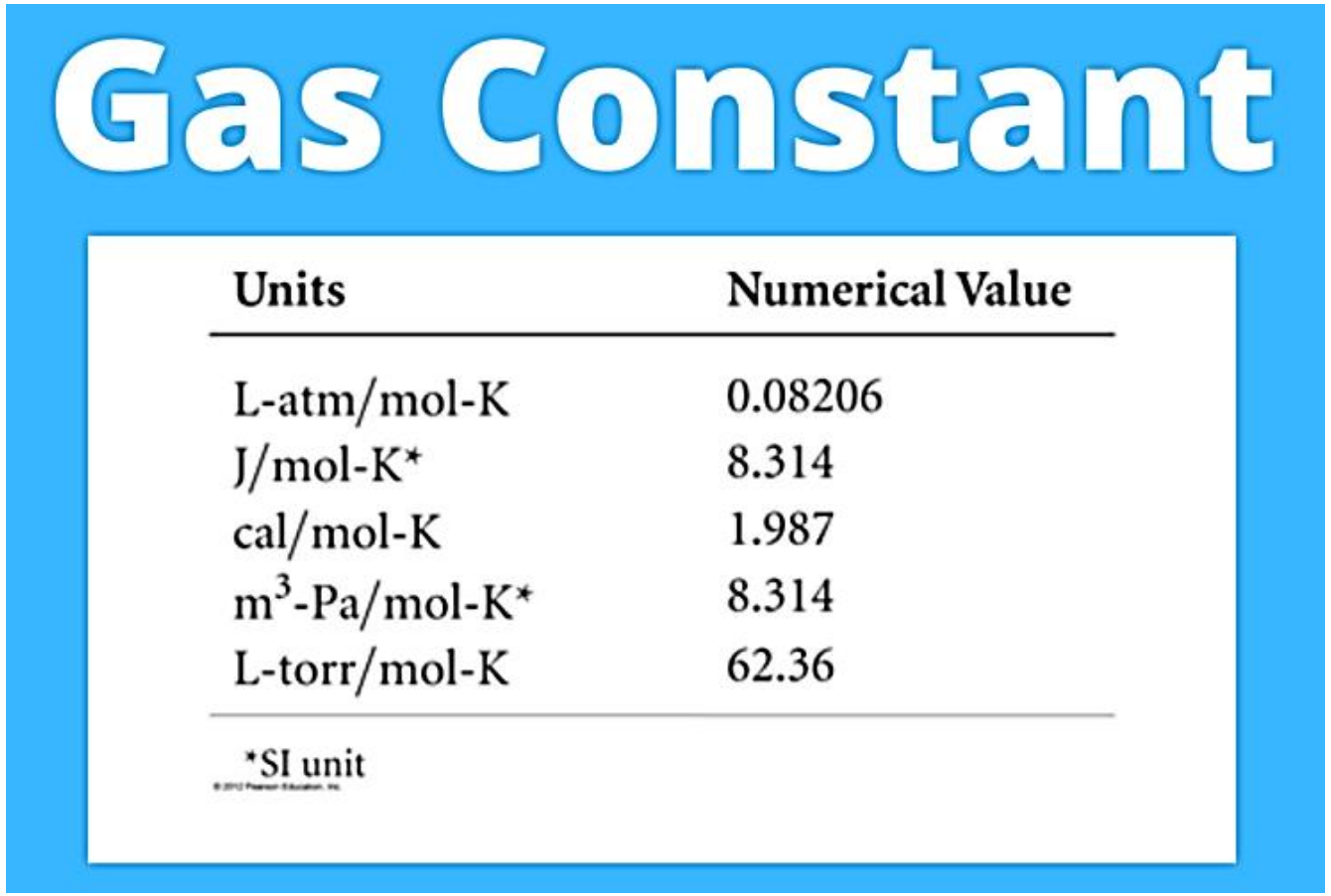
P: absolute pressure

n: number of moles of gas

T: temperature (in kelvins)

R: ideal/universal gas constant

Ideal gas law



Question:

What is the value (and unit) of the gas constant?

Ideal gas law (general gas equation)

$$PV = NkT$$

***k*: Boltzmann constant**

$$k = \frac{R}{N_A} = \frac{8.31 \frac{J}{mol.K}}{6.02\ x\ 10^{23} mol^{-1}}$$

$$k = 1.38\ x\ 10^{-23} J/K$$

Ideal gas law



a)



b)

Courtesy www.doctorslime.com

- A stainless-steel tank was filled with steam of 110°C. Water was poured onto the tank to rapidly cool it. Within less than a minute, the tank was crushed. **Why?**
- The condensation of steam decreased the number N of gas molecules and the temperature T of the gas inside the tank.
- V constant → P decreased (ideal gas law) so much that the external **atmospheric pressure** was able to crush the tank's steel wall.

Work done by an ideal gas

$$P = \frac{nRT}{V} = (a\ constant)\frac{1}{V}$$

$$\mathrm{W} = \int_{V_i}^{V_f} P dV = \int_{V_i}^{V_f} \frac{nRT}{V} dV$$

$$\mathrm{W} = nRT \int_{V_i}^{V_f} \frac{dV}{V} = nRTln \Big|_{V_i}^{V_f}$$

$$\mathrm{W} = nRTln \frac{V_f}{V_i}$$

At constant temperature

Consider an **isothermal expansion** (the reverse is called an **isothermal compression**)

Work done Ideal gas law

$$W = 0$$

At constant volume

Consider an **isochoric process**

Work done Ideal gas law

$$W = P(V_f - V_i) = P\Delta V$$

At constant pressure

Consider an **isobaric process**

Work done Ideal gas law

$$\mathrm{W} = P\left(V_f - V_i\right) = P\Delta V$$

At constant pressure

Consider an **isobaric process**

Exercise

1. Which of the following term does not involve in ideal gas law?
   a) Pressure
   b) Volume
   c) Temperature
   d) Time

Explanation: Ideal gas law is PV = nRT, which does not involve time.

2. Which of the following is not true about ideal gas molecules?
   a) They have negligible size
   b) They do not have attractive forces
   c) They do not apply pressure
   d) They move in random motion

Explanation: Ideal gas molecules apply pressure to the walls of a container.



Exercise

A cylinder contains 12 L of oxygen at 20°C and 15 atm. The temperature is raised to 35°C, and the volume is reduced to 8.5 L. What is the final pressure of the gas in atmospheres? Assume that the gas is ideal.

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Exercise

A cylinder contains 12 L of oxygen at 20°C and 15 atm. The temperature is raised to 35°C, and the volume is reduced to 8.5 L. What is the final pressure of the gas in atmospheres? Assume that the gas is ideal.

$$P_i V_i = nRT_i \; and \; P_f V_f = nRT_f$$

$$\rightarrow P_f = \frac{P_i T_f V_i}{T_i V_f}$$

$$= \frac{15atm \; x \; (273 + 35)K \; x \; 12L}{(273 + 20)K \; x \; 8.5L} = 22atm$$

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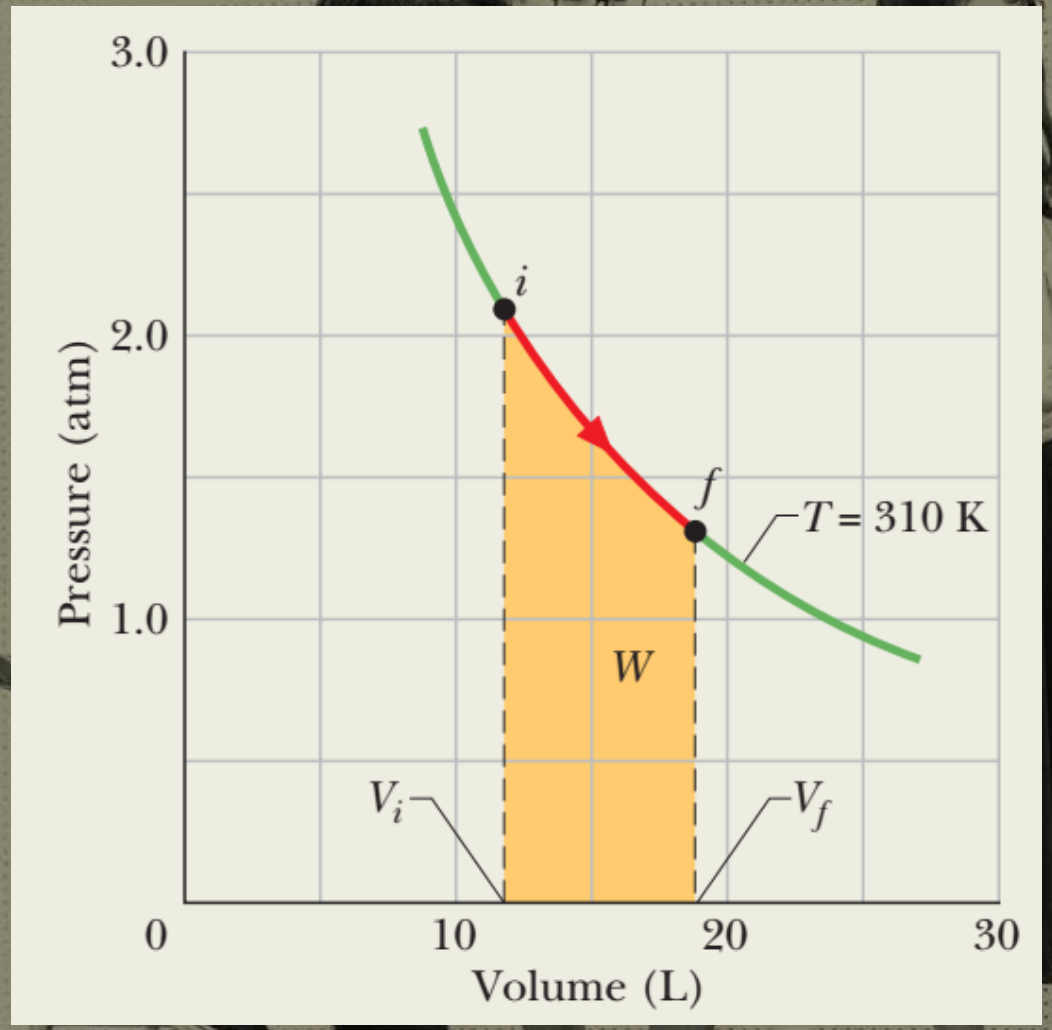
Exercise

One mole of oxygen (assume it to be an ideal gas) expands at a constant temperature T of 310 K from an initial volume V_i of 12 L to a final volume V_f of 19 L.

a) Establish the PV diagram

b) How much work is done by the gas during the expansion?

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Exercise

One mole of oxygen (assume it to be an ideal gas) expands at a constant temperature T of 310 K from an initial volume V_i of 12 L to a final volume V_f of 19 L.

a) Establish the PV diagram

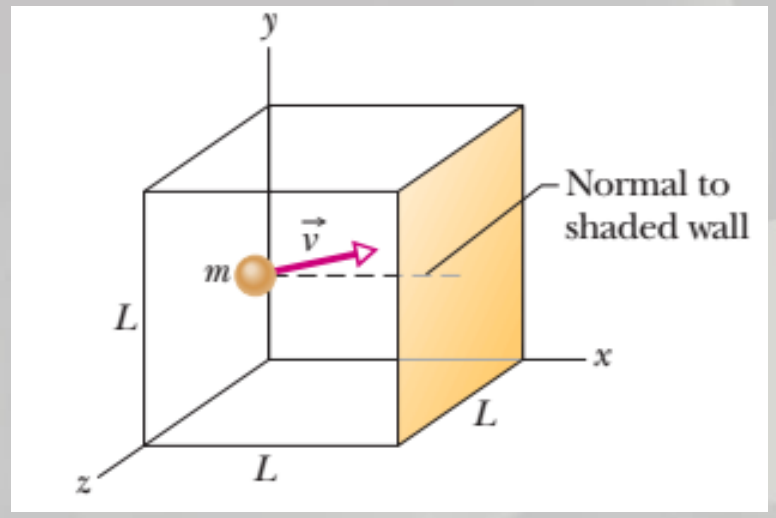
b) How much work is done by the gas during the expansion?

$$W = nRTln\frac{V_f}{V_i}$$

$$= 1mol\ x\ 8.31\frac{J}{mol.K}\ x\ 310K\ x\ \ln\frac{19L}{12L}$$

$$= 1180J$$

Pressure, Temperature, and RMS Speed

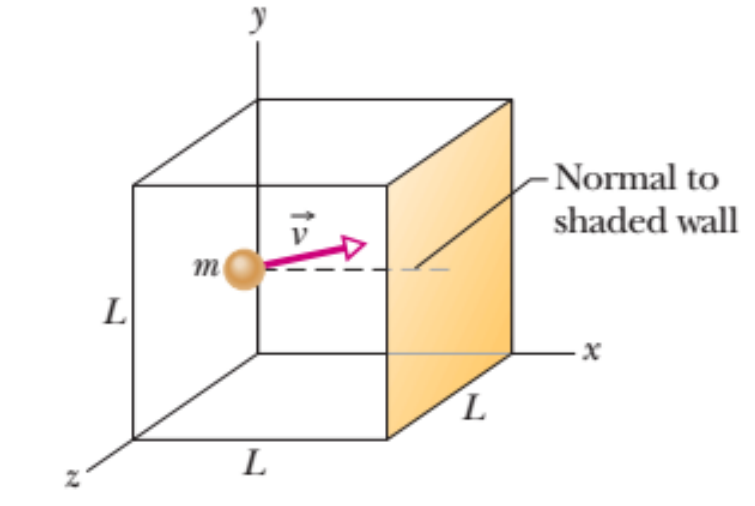


A cubical box of edge length L, containing n moles of an ideal gas. A molecule of mass m and velocity is about to collide with the shaded wall of area L^2 and temperature T.

- We ignore (for the time being) collisions of the molecules with one another and consider only elastic collisions with the walls

What is the connection between the pressure P exerted by the gas on the walls and the speeds of the molecules?

Pressure, Temperature, and RMS Speed



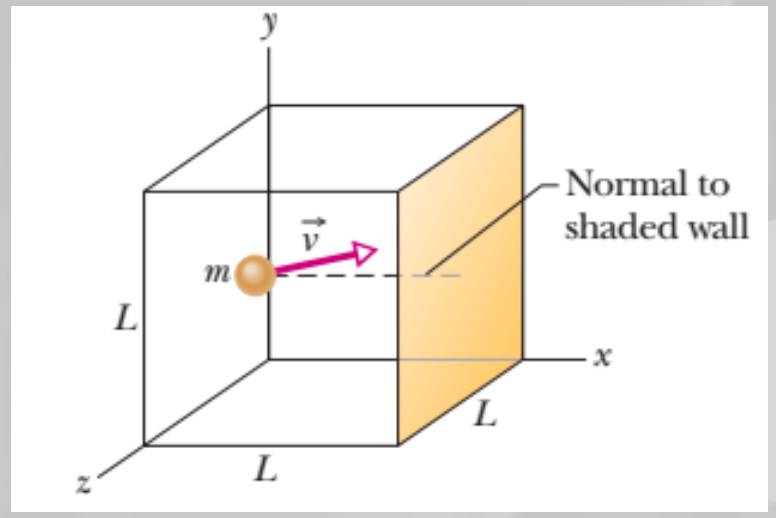
A cubical box of edge length L, containing n moles of an ideal gas. A molecule of mass m and velocity is about to collide with the shaded wall of area L^2 and temperature T.

- When this molecule collides with the shaded wall, the only component of its velocity that is changed is the x component, and that component is reversed (because we consider only elastic collisions)
- The only change in the **particle's momentum** is along the x axis:

$$\Delta p_x = (-mv_x) - (mv_x) = -2mv_x$$

- Hence, the momentum Δp_x delivered to the wall by the molecule during the collision is $+2mv_x$

Pressure, Temperature, and RMS Speed



A cubical box of edge length L, containing n moles of an ideal gas. A molecule of mass m and velocity is about to collide with the shaded wall of area L^2 and temperature T.

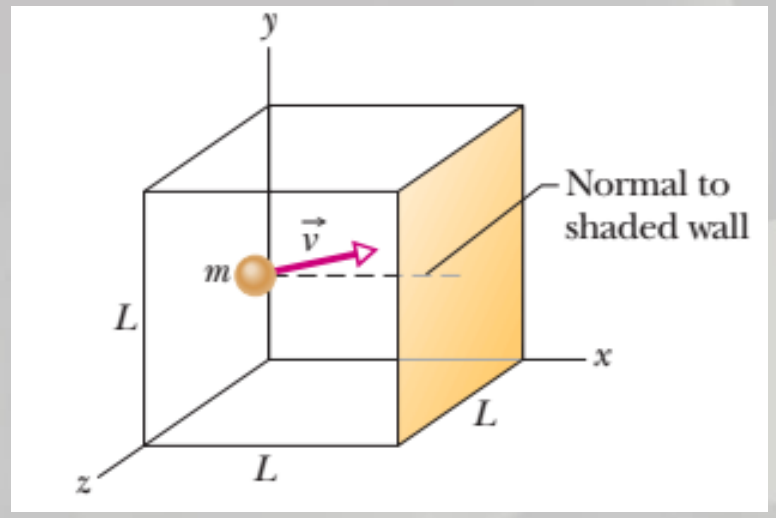
- The time Δt between collisions is the time the molecule takes to travel to the opposite wall and back again (a distance $2L$) at speed v_x

$$\Delta t = \frac{2L}{v_x}$$

- The average rate at which momentum is delivered to the shaded wall by this single molecule is

$$\frac{\Delta p_x}{\Delta t} = \frac{2mv_x}{2L/v_x} = \frac{mv_x{}^2}{L}$$

Pressure, Temperature, and RMS Speed

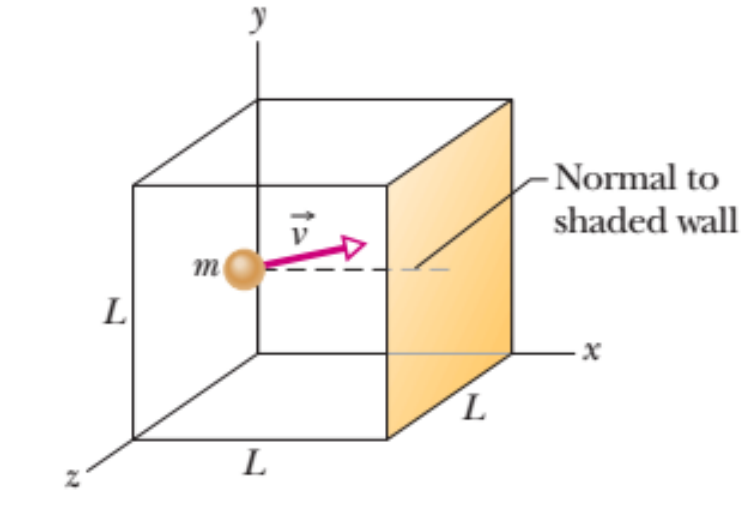


A cubical box of edge length L, containing n moles of an ideal gas. A molecule of mass m and velocity is about to collide with the shaded wall of area L^2 and temperature T.

- From Newton's second law ($\vec{F} = d\vec{p}/dt$), the rate at which momentum is delivered to the wall is the force acting on that wall.
- Pressure P = Total force F_x (of all molecules of different speeds)/area (L^2)

$$P = \frac{F_x}{L^2} = \frac{\frac{m{v_{x1}}^2}{L} + \cdots + \frac{m{v_{xN}}^2}{L}}{L^2}$$
$$= \frac{m}{L^3}({v_{x1}}^2 + \ldots {v_{xN}}^2)$$

Pressure, Temperature, and RMS Speed



A cubical box of edge length *L*, containing *n* moles of an ideal gas. A molecule of mass *m* and velocity is about to collide with the shaded wall of area L^2 and temperature T.

- By replacing: $N = nN_A$

  $(v_{x1}{}^2 + \ldots v_{xN}{}^2) = nN_A(v_x)^2{}_{avg}$

- Then:

$$P = \frac{mnN_A}{L^3}(v_x)^2{}_{avg}$$

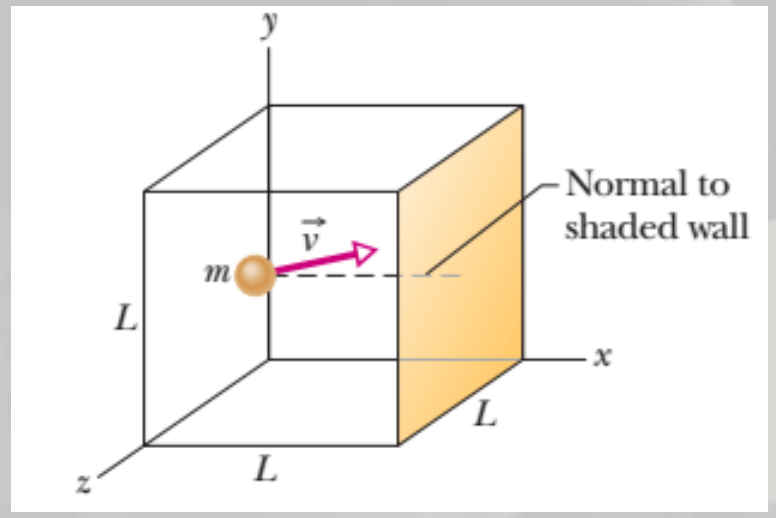
- By replacing: $mN_A = M\ (molar\ mass)$

  $L^3 = V\ (volume)$

- Then:

$$P = \frac{nM}{V}(v_x)^2{}_{avg}$$

Pressure, Temperature, and RMS Speed



A cubical box of edge length *L*, containing *n* moles of an ideal gas. A molecule of mass *m* and velocity is about to collide with the shaded wall of area L^2 and temperature T.

- For any molecule:

$$v^2 = v_x^2 + v_y^2 + v_z^2$$

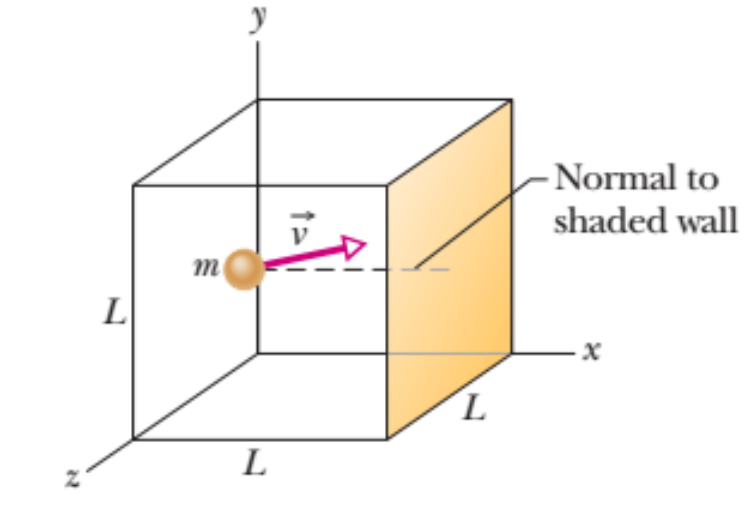
- Because molecules are all moving in random directions:

$$v_x^2 = v_y^2 = v_z^2 = \frac{1}{3}v^2$$

- Then:

$$P = \frac{nM}{3V}(v^2)_{avg}$$

Pressure, Temperature, and RMS Speed



A cubical box of edge length L, containing n moles of an ideal gas. A molecule of mass m and velocity is about to collide with the shaded wall of area L^2 and temperature T.

- We have:

$$\sqrt{(v^2)_{avg}} = v_{rms} \text{ (root-mean square speed)}$$

- Then:

$$P = \frac{nM}{3V}(v^2)_{rms}$$

- Combining with ideal gas law ($PV = nRT$):

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

Some RMS Speeds at Room Temperature ($T = 300$ K)[a]

Gas	Molar Mass (10^{-3} kg/mol)	v_{rms} (m/s)
Hydrogen (H_2)	2.02	1920
Helium (He)	4.0	1370
Water vapor (H_2O)	18.0	645
Nitrogen (N_2)	28.0	517
Oxygen (O_2)	32.0	483
Carbon dioxide (CO_2)	44.0	412
Sulfur dioxide (SO_2)	64.1	342

[a]For convenience, we often set room temperature equal to 300 K even though (at 27°C or 81°F) that represents a fairly warm room.

- $P = \frac{nM}{3V}(v^2)_{rms}$: how the pressure of the gas (a purely macroscopic quantity) depends on the speed of the molecules (a purely microscopic quantity)
- $v^2{}_{rms}$: only a kind of average speed; many molecules move much faster than this, and some much slower.

Why root-mean-square speed and not average speed?

- **Kinetic Energy Relation**: The kinetic energy of a gas particle is proportional to the square of its velocity. The average speed doesn't account for this squared relationship, so doesn't provide a direct connection to the total kinetic energy.
- **Direction and Velocity Components:** In gases, particles move randomly in all directions. When calculating average velocity, some velocities could cancel out because of their direction. The rms speed overcomes this issue by squaring the velocities (making them positive), taking their average and then square root.
- **Statistical Distribution**: The distribution of speeds in a gas follows the Maxwell-Boltzmann distribution, where most particles have intermediate speeds, and only a few have very low or very high speeds. The rms speed gives more weight to the higher speeds, which is statistically significant when understanding the system's energy distribution.



Exercise

Here are five numbers: 5, 11, 32, 67, and 89.
(a) What is the average value n_{avg} of these numbers?

(b) What is the rms value n_{rms} of these numbers?



Exercise

Here are five numbers: 5, 11, 32, 67, and 89.

(a) What is the average value n_{avg} of these numbers?

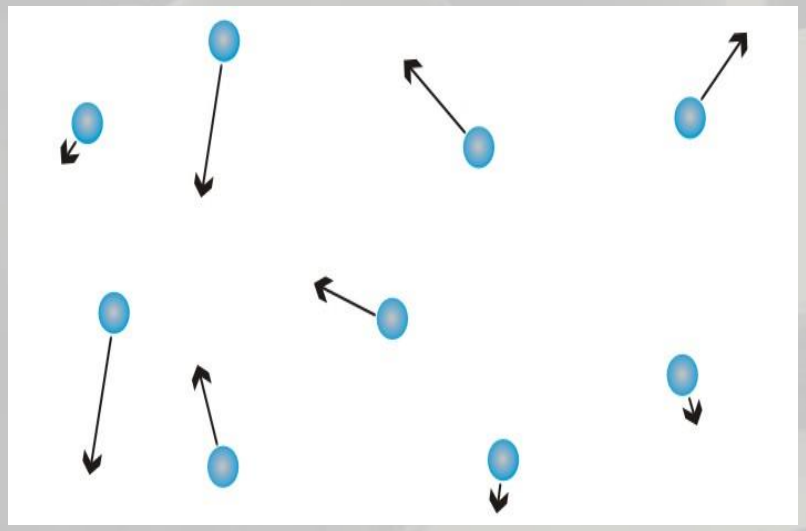
$$n_{avg} = \frac{5 + 11 + 32 + 67 + 89}{5} = 40.8$$

(b) What is the rms value n_{rms} of these numbers?

$$\mathrm{n}_{rms} = \sqrt{\frac{5^2 + 11^2 + 32^2 + 67^2 + 89^2}{5}}$$

$$= 52.1$$

Translational kinetic energy



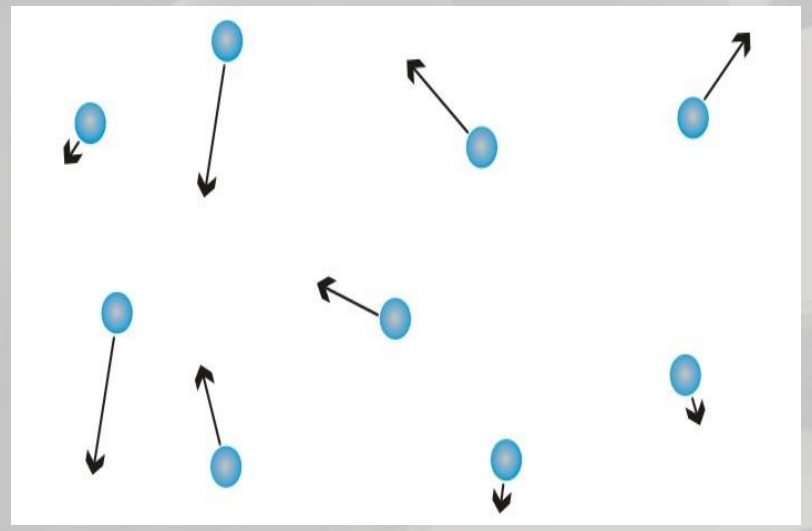
- Consider a single molecule of an ideal gas having speed changes when it collides with other molecules
- Its translational kinetic energy at any instant:

$$K = \frac{1}{2}mv^2$$

- Its *average* translational kinetic energy over time:

$$K_{avg} = \frac{1}{2}m(v^2)_{avg} = \frac{1}{2}mv^2{}_{rms} = \frac{1}{2}m\frac{3RT}{M}$$

Translational kinetic energy



- Substituting:

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

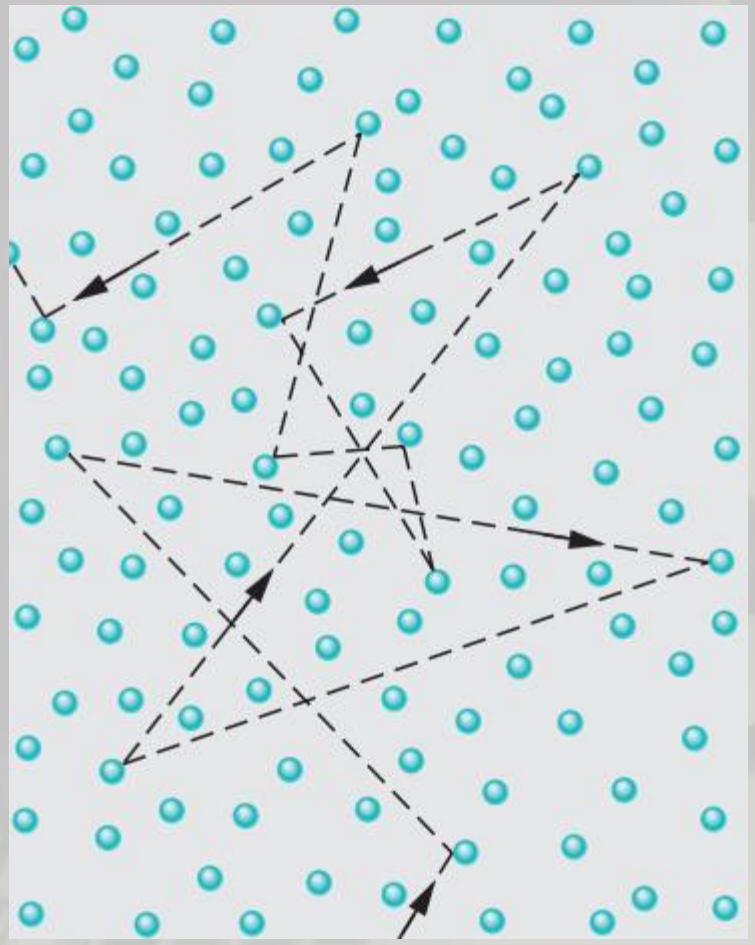
$$N_A = M/m$$

- Thus:

$$K_{avg} = \frac{3}{2}\frac{RT}{N_A} = \frac{3}{2}kT$$

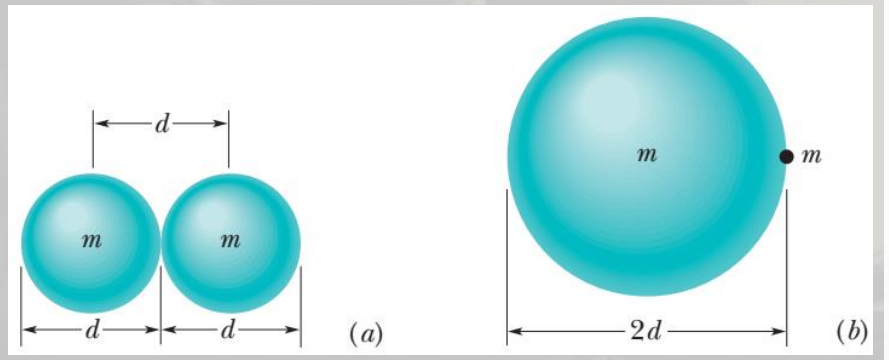
At a given temperature T, all ideal gas molecules - no matter what their mass - have the same average translational kinetic energy

Mean free path



- A typical molecule moves through the gas, changing both speed and direction abruptly as it collides elastically with other molecules.
- Between collisions, the molecule moves in a straight line at constant speed.
- **Mean free path λ**: the average distance traversed by a molecule between collisions.

Mean free path

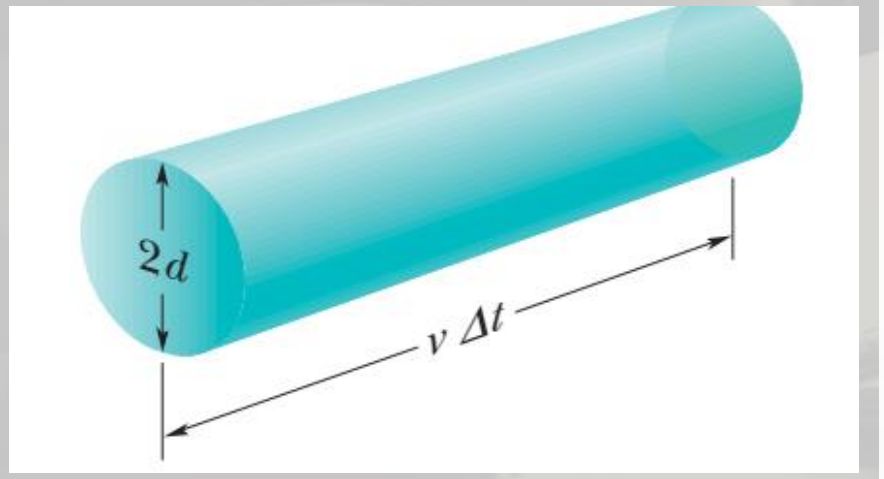


- Consider a single sphere molecule (diameter *d*) traveling with *v constant,* other molecules at rest.

(a) A collision occurs when the centers of two molecules come within a distance *d* of each other, *d* being the molecular diameter.

(b) An equivalent representation is to think of the moving molecule as having a *radius d* and all other molecules as being points. The condition for a collision is unchanged.

Mean free path

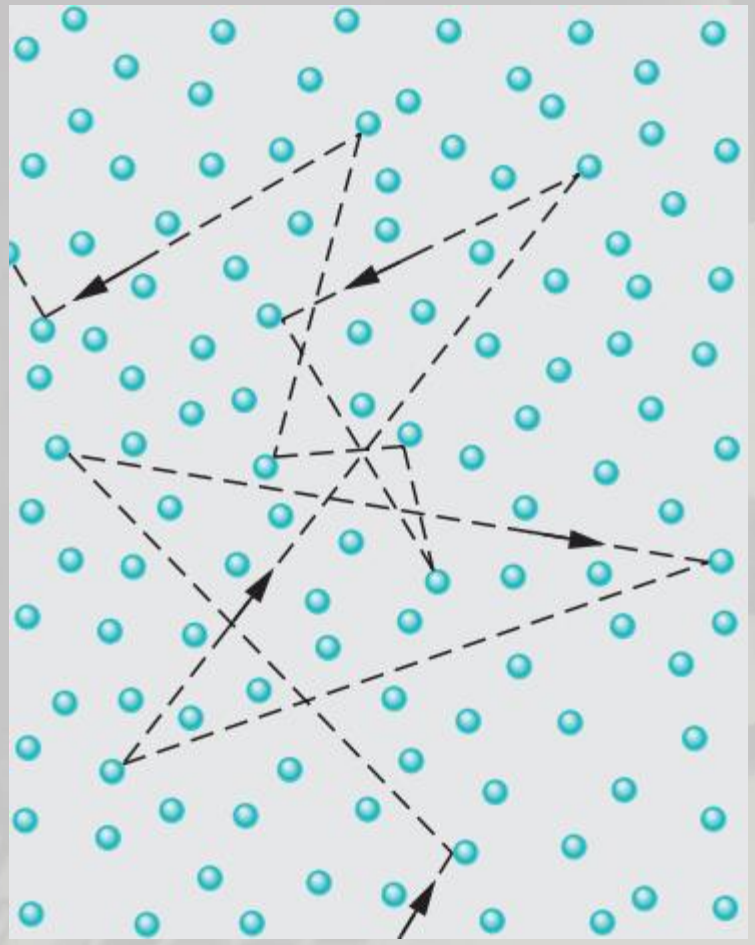


- In time Δt the moving molecule sweeps out a cylinder of length $v\Delta t$ and radius d and volume $(\pi d^2)(v\Delta t)$
- The number of collisions that occur in time Δt is then equal to the number of (point) molecules that lie within this cylinder.
- Number of molecules in the cylinder

= number of molecules per unit volume x volume

$$= N/V \; x \; (\pi d^2)(v\Delta t)$$

Mean free path



$$\lambda = \frac{length\ of\ path\ during\ \Delta t}{number\ of\ collisions\ in\ \Delta t}$$

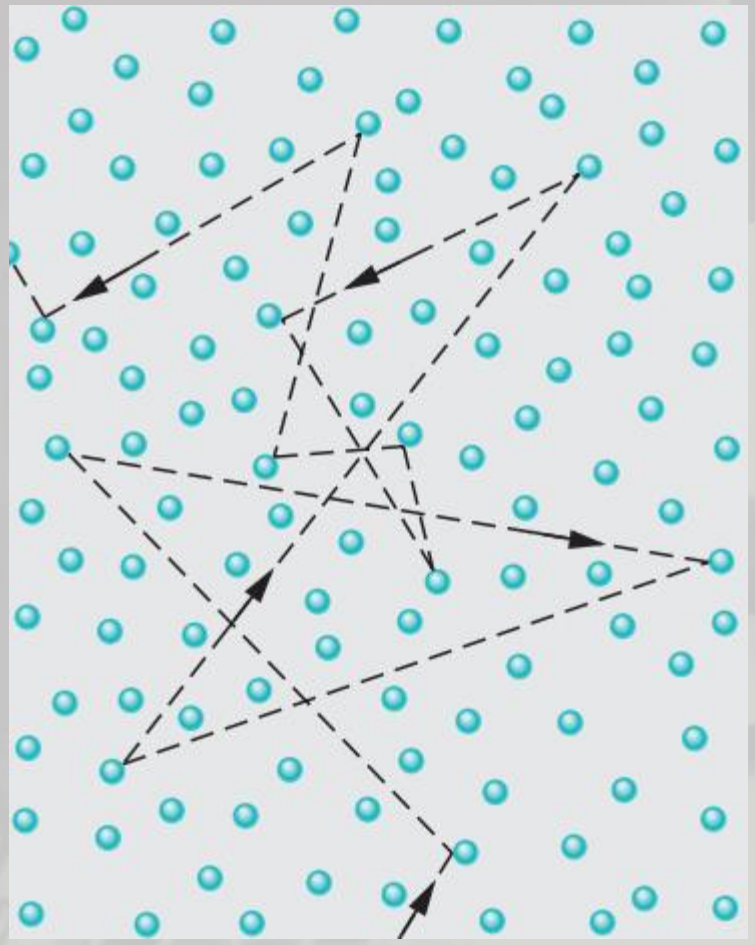
$$\lambda \approx \frac{v\Delta t}{(\pi d^2)(v\Delta t)N/V}$$

$$= \frac{1}{\pi d^2 N/V}$$

- **Be careful:** This equation is based on the assumption that all the molecules except one are at rest.

In fact, *all* the molecules are moving!!!

Mean free path



$$\lambda = \frac{length\ of\ path\ during\ \Delta t}{number\ of\ collisions\ in\ \Delta t}$$

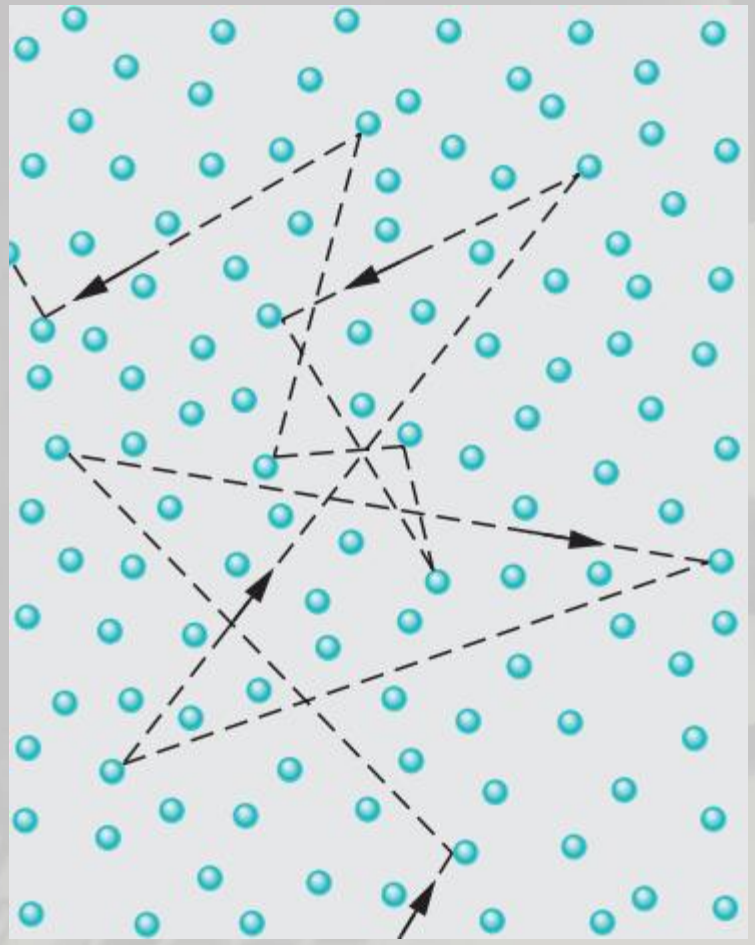
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Mean free path

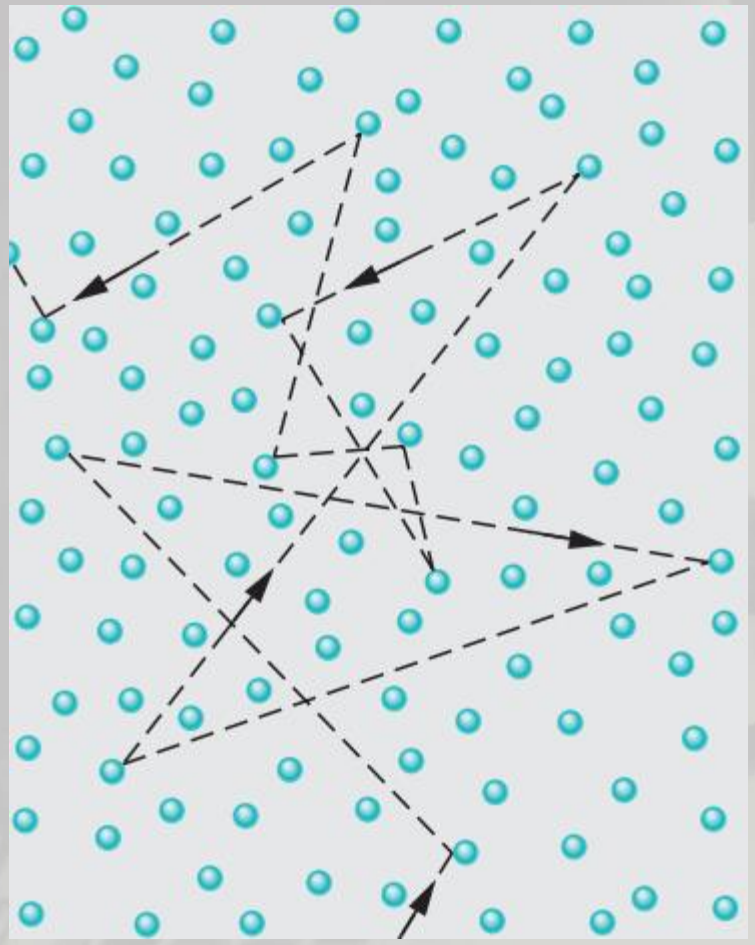


$$\lambda \approx \frac{v_{avg}\Delta t}{(\pi d^2)(v_{rel}\Delta t)N/V} = \frac{1}{\pi d^2 N/V}$$

- We have canceled two v symbols:

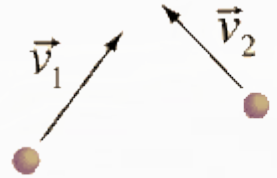
- The v_{avg} in the numerator: the mean speed of the molecules *relative to the container*.
- The v_{rel} in the denominator: the mean speed of our single molecule *relative to the other molecules*, which are moving.

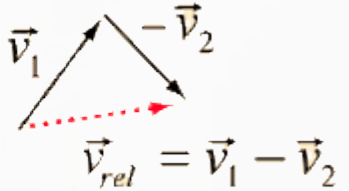
Mean free path



- A recall of mathematics….

Random molecular velocities





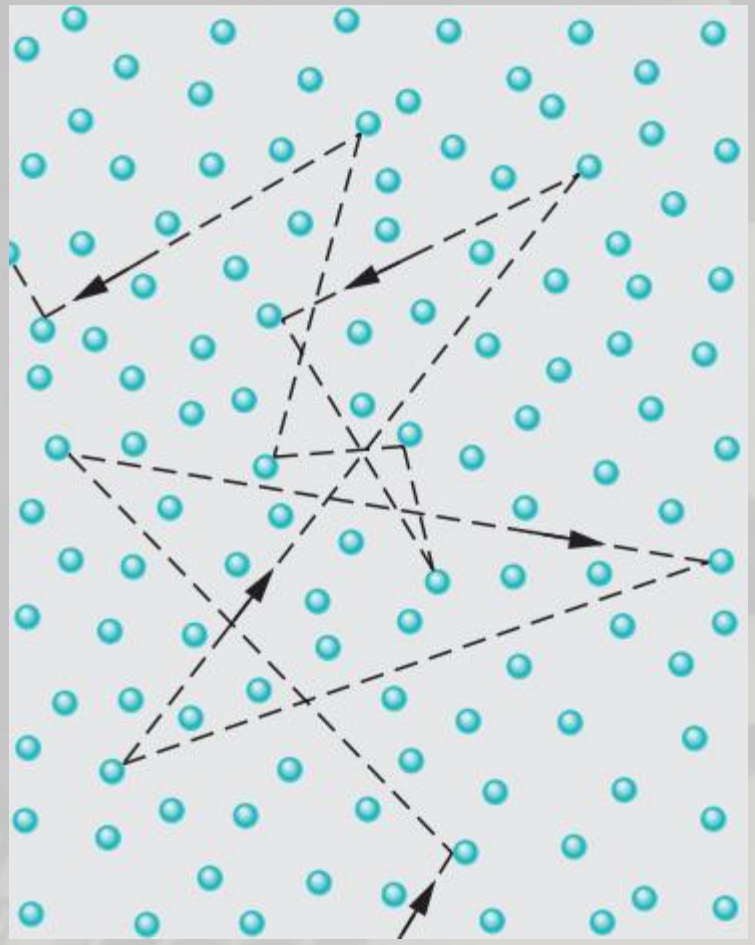
$$\overline{v_{rel}} = \sqrt{\overline{\vec{v}_1 \cdot \vec{v}_1} - 2\overline{\vec{v}_1 \cdot \vec{v}_2} + \overline{\vec{v}_2 \cdot \vec{v}_2}}$$

$= 0$ since $\vec{v}_1$ and $\vec{v}_2$ are random and uncorrelated.

$$\overline{v_{rel}} = \sqrt{\overline{v}_1^2 + \overline{v}_2^2}$$

- Since $\overline{v_1} = \overline{v_2} \rightarrow \overline{v_{rel}} = \sqrt{2}\bar{v}$

Mean free path



- Let's re-write our formula:

$$\lambda = \frac{\bar{v}\Delta t}{(\pi d^2)(\overline{v_{rel}}\Delta t)N/V}$$

$$= \frac{1}{\sqrt{2}\pi d^2 N/V}$$

- Air molecules:

  At sea level: λ = 0.1 mm.

  At h = 100 km, λ = 16 cm.

  At h = 300 km, λ = 20 km.



Exercise

a) What is the mean free path λ for oxygen molecules at temperature $T = 300$ K and pressure $p = 1.0$ atm? Assume that the molecular diameter is $d = 290$ pm and the gas is ideal.

(b) Assume the average speed of the oxygen molecules is $v = 450$ m/s. What is the average time t between successive collisions for any given molecule?



Exercise

a) What is the mean free path λ for oxygen molecules at temperature $T = 300$ K and pressure $P = 1.0$ atm $= (10^5$ Pa)? Assume that the molecular diameter is $d = 290$ pm and the gas is ideal.

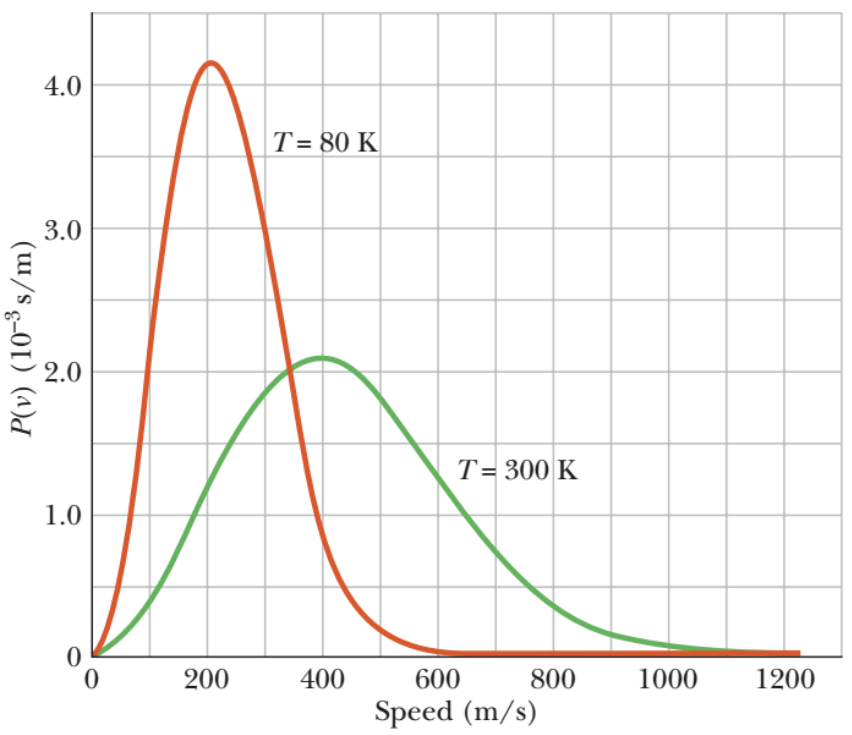
- We first need the number of molecules per unit volume, N/V. Assuming an ideal gas:

$$\lambda = \frac{1}{\sqrt{2}\pi d^2 N/V} = \frac{kT}{\sqrt{2}\pi d^2 P} = 1.1\ x\ 10^{-7}\ m$$

(b) Assume the average speed of the oxygen molecules is $v =$ 450 m/s. What is the average time t between successive collisions for any given molecule?

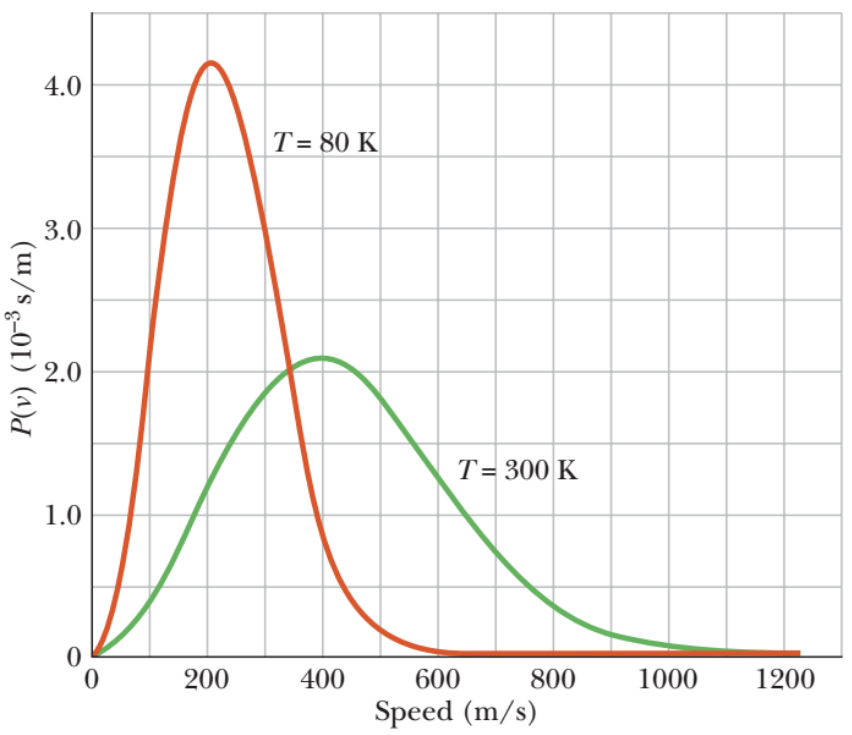
$$t = \frac{distance}{speed} = \frac{\lambda}{v} = 2.44\ x\ 10^{-10}\text{s}$$

The distribution of molecular speeds



- What fraction of the molecules have speeds greater than the rms value?
- We need to know how the possible values of speed are distributed among the molecules to answer the above question
- In 1852, Scottish physicist James Clerk Maxwell first solved the problem of finding the speed distribution of gas molecules

The distribution of molecular speeds

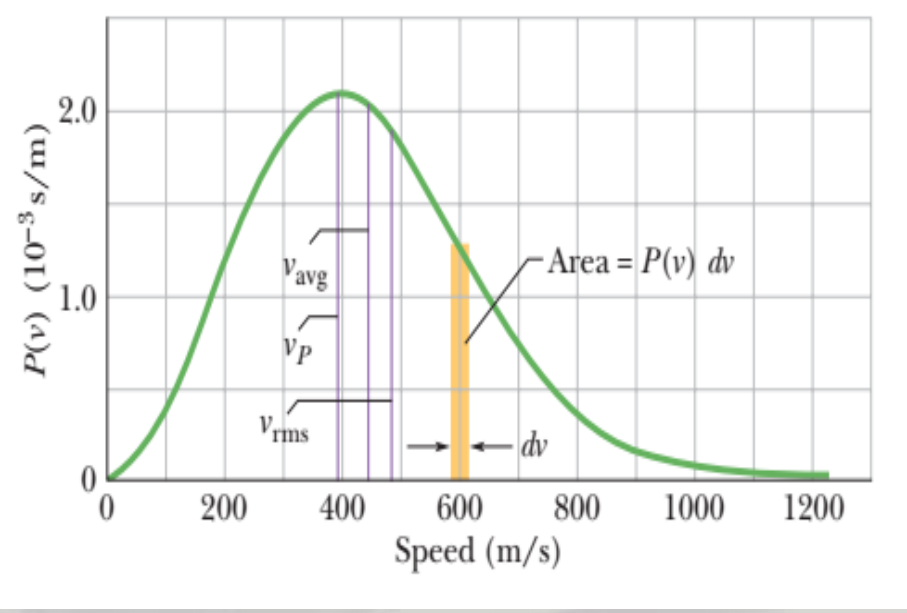


- **Maxwell's speed distribution law:**

$$P(v) = 4\pi\left(\frac{M}{2\pi RT}\right)^{3/2} v^2 e^{-Mv^2/2RT}$$

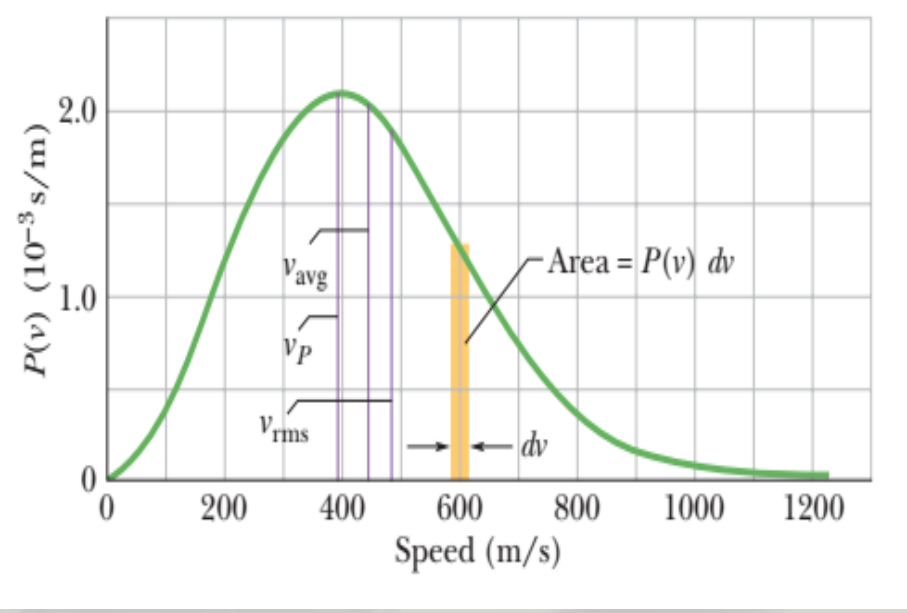
P(v): probability distribution function
M: Molar mass
T: Temperature
v: molecular speed
R: gas constant

The distribution of molecular speeds



- $P(v)$ dv: fraction of molecules with speeds in the interval dv centered on speed v
- The total area under the distribution curve corresponds to the fraction of the molecules whose speeds lie between zero and infinity, having value: $\int_0^\infty P(v)\,dv = 1$
- The fraction of molecules with speeds in an interval of v_1 to v_2 is then: $\text{frac} = \int_{v_1}^{v_2} P(v)\,dv$

Average, RMS, and Most probable speeds



- Average speed: $v_{avg} = \sqrt{\frac{8RT}{\pi M}}$
- RMS speed: $v_{rms} = \sqrt{\frac{3RT}{M}}$
- Most probable speed: $v_P = \sqrt{\frac{2RT}{M}}$



Exercise

In oxygen (molar mass $M = 0.0320$ kg/mol) at room temperature (300 K)

(a) What is the average speed v_{avg} of oxygen gas molecules at $T = 300$ K?

(b) What is the root-mean-square speed v_{rms} at 300 K?

(c) What is the most probable speed v_P at 300 K?



Exercise

For oxygen gas molecules (molar mass $M = 0.0320$ kg/mol) at room temperature (300 K)

(a) What is v_{avg} at $T = 300$ K?

$$v_{avg} = \sqrt{\frac{8RT}{\pi M}}$$
$$= \sqrt{\frac{8(8.31\ \text{J/mol}\cdot\text{K})(300\ \text{K})}{\pi(0.0320\ \text{kg/mol})}}$$
$$= 445\ \text{m/s}.$$

(b) What is v_{rms} at 300 K?

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$
$$= \sqrt{\frac{3(8.31\ \text{J/mol}\cdot\text{K})(300\ \text{K})}{0.0320\ \text{kg/mol}}}$$
$$= 483\ \text{m/s}.$$

(c) What is v_P at 300 K?

$$v_P = \sqrt{\frac{2RT}{M}}$$
$$= \sqrt{\frac{2(8.31\ \text{J/mol}\cdot\text{K})(300\ \text{K})}{0.0320\ \text{kg/mol}}}$$
$$= 395\ \text{m/s}.$$

Molar specific heats of an ideal gas



- In this part, we want to derive from molecular considerations an expression for the internal energy E_{int} of an ideal gas.
- In other words, we want an expression for the energy associated with the random motions of the atoms or molecules in the gas.
- We shall then use that expression to derive the molar specific heats of an ideal gas.

Internal energy of an ideal gas

Internal Energy of a system is the sum of kinetic energies of all of its constituent particles, plus the sum of all the potential energies of interaction among these particles.

(Remember: potential energies of interaction among ideal gas particles are considered zero!!

- Considering a *monatomic ideal gas* (individual atoms rather than molecules: He, Ar …)
- Assuming E_{int} = sum of the translational kinetic energies of the atoms (Quantum theory disallows rotational kinetic energy for individual atoms.)
- Average translational kinetic energy of a single atom: $K_{avg} = \frac{3}{2}kT = E_{int}$
- For sample of n moles: $E_{int} = nN_A\frac{3}{2}kT$
- With $k = R/N_A$:

$$E_{int} = \frac{3}{2}nRT\ (monatomic\ ideal\ gas)$$

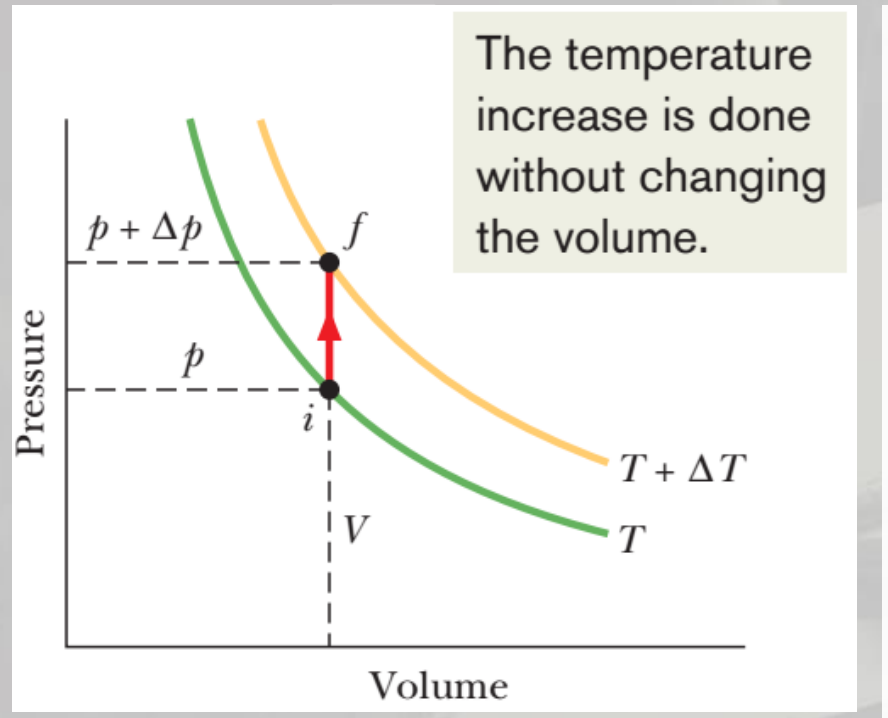
Internal energy of an ideal gas

Internal Energy of a system is the sum of kinetic energies of all of its constituent particles, plus the sum of all the potential energies of interaction among these particles.

(Remember: potential energies of interaction among ideal gas particles are considered zero!!

- We are now able to derive two expressions:
- **Molar specific heat and constant volume** (*Cv*), for the case in which the volume of the gas remains constant as energy is transferred to or from it as heat.
- **Molar specific heat and constant pressure** (*Cp*), for the case in which the pressure of the gas remains constant as energy is transferred to or from it as heat.
- By convention, the capital letter *C* is used even though *Cv* and *Cp* represent types of specific heat and not heat capacities.

Molar specific heat at constant volume (C_v)

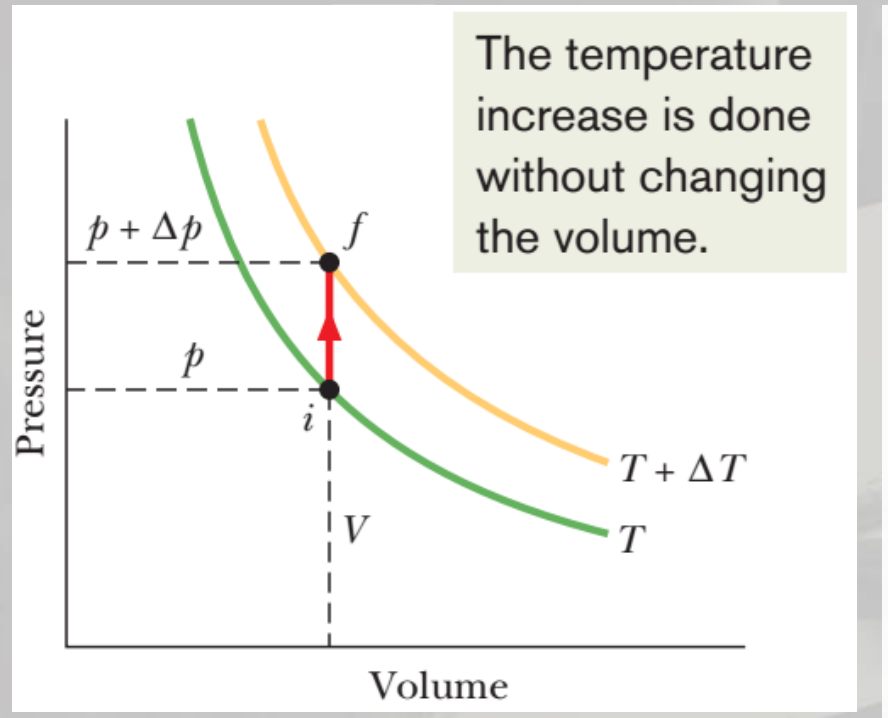


The temperature of an ideal gas is raised from T to $T+\Delta T$ in a constant volume process

- Considering n moles of an ideal gas at P and T, confined to a fixed V
- The gas temperature rises from T to $T+\Delta T$, and its pressure rises from P to $P+\Delta P$
- The heat Q is related to the temperature change ΔT by:

$$Q = nC_v \Delta T$$

Molar specific heat at constant volume (C_v)



The temperature of an ideal gas is raised from T to $T+\Delta T$ in a constant volume process

- Substituting $Q = nC_v \Delta T$ in the first law:

$$\Delta E_{int} = nC_v \Delta T - W$$

- With $V = const$, $W = 0$:

$$\Delta E_{int} = nC_v \Delta T$$

- Then:

$$C_v = \frac{\Delta E_{int}}{n \Delta T}$$

- Substituting $\Delta E_{int} = \frac{3}{2} nR \Delta T$:

$$C_v = \frac{3}{2} R = 12.5\ J/mol.K\ (monatomic\ gas)$$

Molar specific heat at constant volume (C_v)

Molecule	Example	C_V (J/mol·K)
Monatomic	Ideal	$\frac{3}{2}R = 12.5$
	Real	He 12.5
		Ar 12.6
Diatomic	Ideal	$\frac{5}{2}R = 20.8$
	Real	N_2 20.7
		O_2 20.8
Polyatomic	Ideal	$3R = 24.9$
	Real	NH_4 29.0
		CO_2 29.7

- The values of C_v for *diatomic gases* (having molecules with two atoms) and *polyatomic gases* (having molecules with more than two atoms) are greater than those for monatomic gases (the more complex molecules can rotate and thus have rotational kinetic energy)
- When Q is transferred to a diatomic or polyatomic gas, only part of it goes into the translational kinetic energy, increasing T

Molar specific heat at constant volume (C_v)

Molecule	Example	C_V (J/mol·K)
Monatomic	Ideal	$\frac{3}{2}R = 12.5$
	Real	He 12.5
		Ar 12.6
Diatomic	Ideal	$\frac{5}{2}R = 20.8$
	Real	N_2 20.7
		O_2 20.8
Polyatomic	Ideal	$3R = 24.9$
	Real	NH_4 29.0
		CO_2 29.7

- We can obtain a general equation for E_{int} of any ideal gas by substituting C_v for *3/2R*:

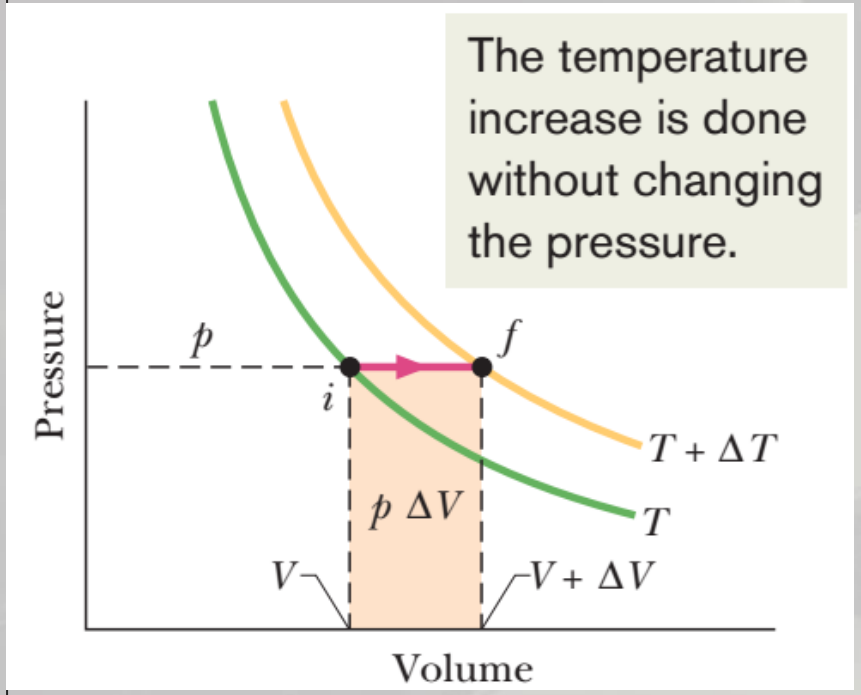
$$E_{int} = nC_vT$$

$$(any\ ideal\ gas: mona - dia - poly)$$

$$\Delta E_{int} = nC_v\Delta T\ (ideal\ gas, any\ process)$$

A change in the internal energy E_{int} of a confined ideal gas depends on only the change in the temperature, not on what type of process produces the change.

Molar specific heat at constant pressure(C_p)



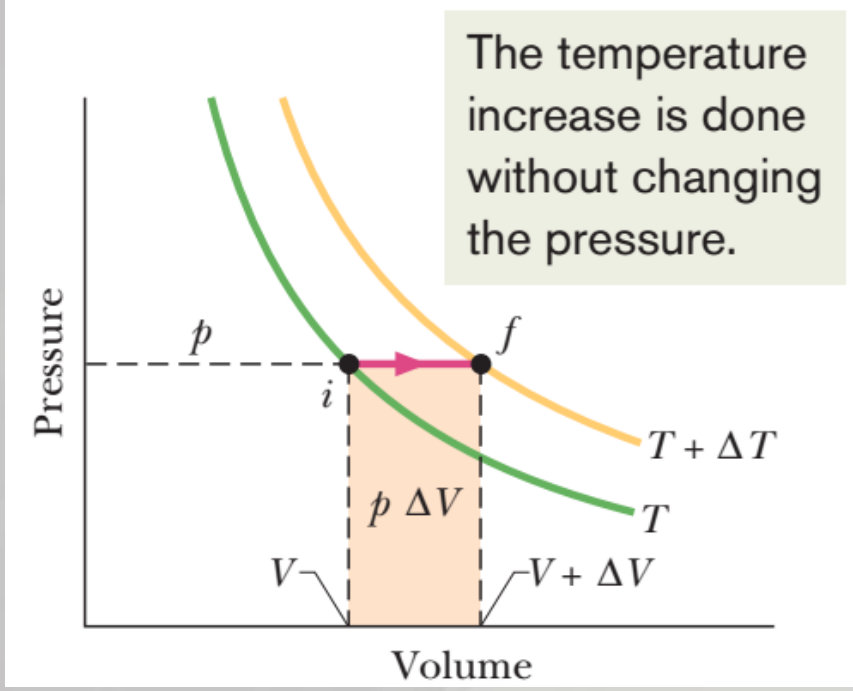
The temperature of an ideal gas is raised from T to $T + \Delta T$ in a constant pressure process. Heat is added and work.

- Considering n moles of an ideal gas at T and V, constant P
- The gas temperature rises to $T + \Delta T$, and its pressure rises to $P + \Delta P$
- The heat Q is related to the temperature change ΔT by:

$$Q = nC_p \Delta T$$

- C_p is *greater* than C_v, because energy must now be supplied not only to raise T but also for the gas to do work

Molar specific heat at constant pressure (C_p)



The temperature of an ideal gas is raised from T to $T + \Delta T$ in a constant pressure process. Heat is added and work.

- To relate molar specific heats C_p and C_v

$$\Delta E_{int} = nC_v\Delta T;\ Q = nC_p\Delta T$$

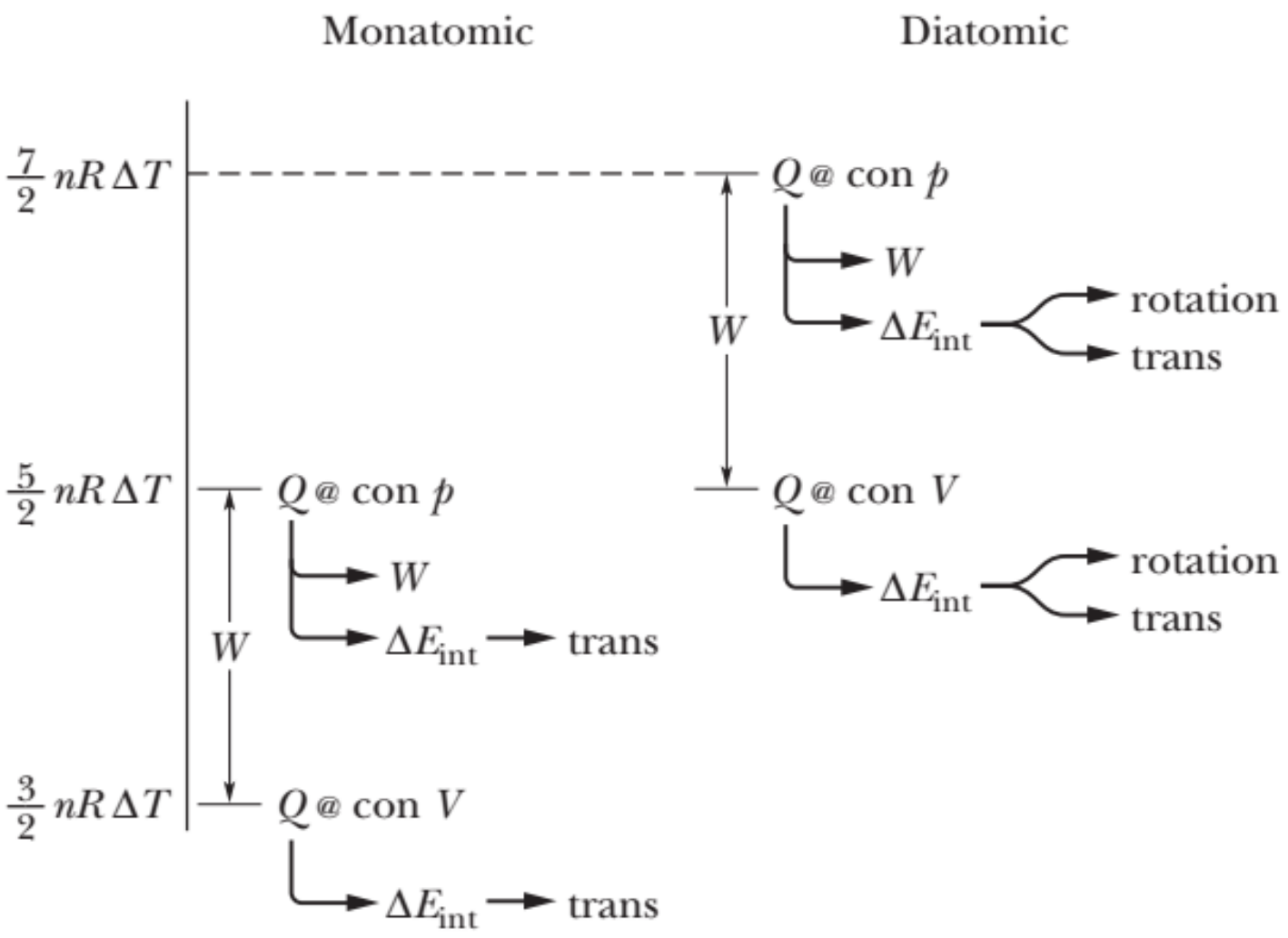
$$PV = nRT \rightarrow W = P\Delta V = nR\Delta T$$

- Then:

$$\Delta E_{int} = Q - W \rightarrow nC_v\Delta T = nC_p\Delta T - nR\Delta T$$

- Finally:

$$C_v = C_p - R \text{ or } C_p = C_v + R$$



The relative values of Q for a monatomic gas and a diatomic gas undergoing a constant-volume process (labeled "con V") and a constant-pressure process (labeled "con p")

Exercise

1. The amount of heat required to raise the temperature of the unit mass of gas through one degree at constant volume is
a) Cv
b) Cp
c) Latent heat
d) none of the mentioned

Answer: A

2. Which one of correct relation ?
a) Cp + Cv = 0
b) Cp = 1 + R/Cv
c) 1 = Cp/Cv
d) Cp = R + Cv

Answer: D

Exercise

3. The heat capacity depends on whether the heat is added at constant _____ or constant _____

a) Volume, Temperature
b) Pressure, Volume
c) Pressure, Temperature
d) Temperature, Pressure

Answer: b

4. Consider He gas an ideal gas, What is Cv for He?

a) 1.5 R
b) 2.5 R
c) 3.5 R
d) 4.5 R

Answer: a



Exercise

A bubble of 5.00 mol of helium is submerged at a certain depth in liquid water when the water (and thus the helium) undergoes a temperature increase T of 20.0°C- at constant pressure. As a result, the bubble expands. The helium is monatomic and ideal.
(a) How much energy is added to the helium as heat during the increase and expansion?

(b) What is the change ΔE_{int} in the internal energy of the helium during the temperature increase?

(c) How much work W is done by the helium as it expands against the pressure of the surrounding water during the temperature increase?



Exercise

$$n = 5\ \text{mol of } He, \Delta T = 20°C, P = const$$

(a) Q?

$$Q = nC_p\Delta T\ (P = \text{const})$$

$$Q = n(C_v + R)\Delta T = n\left(\frac{3}{2}R + R\right)\Delta T = n\frac{5}{2}R\Delta T$$

$$= 2080J$$

(b) ΔE_{int}?

$$\Delta E_{int} = nC_v\Delta T = n\frac{3}{2}R\Delta T = 1250J$$

(c) W?

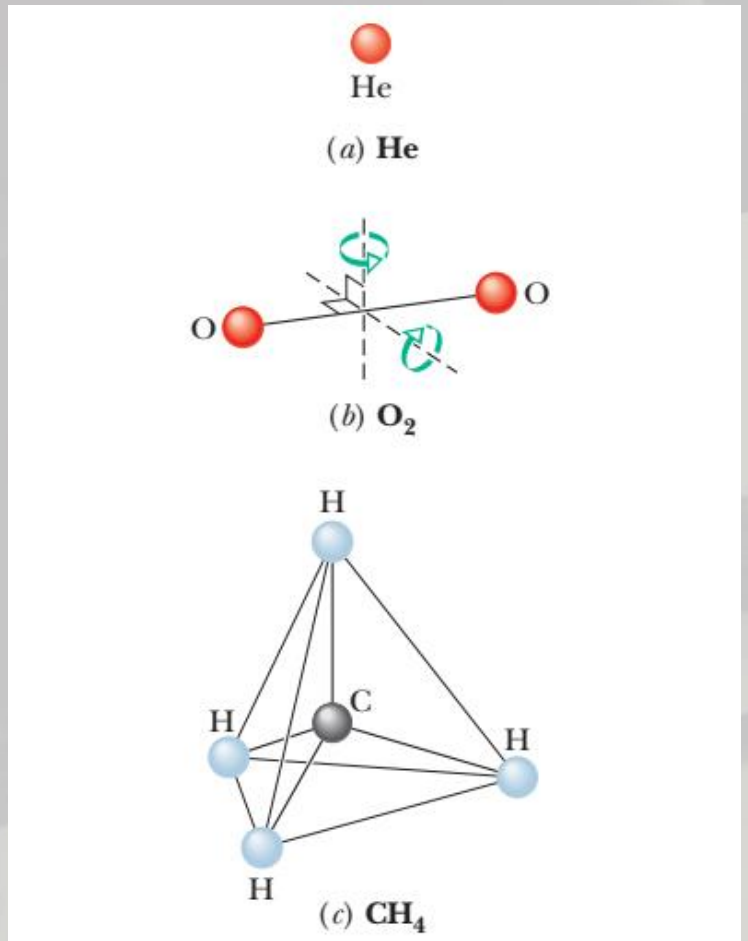
$$PV = nRT \rightarrow W = P\Delta V = nR\Delta T$$

Another way: $W = Q - \Delta E_{int} = 831J$



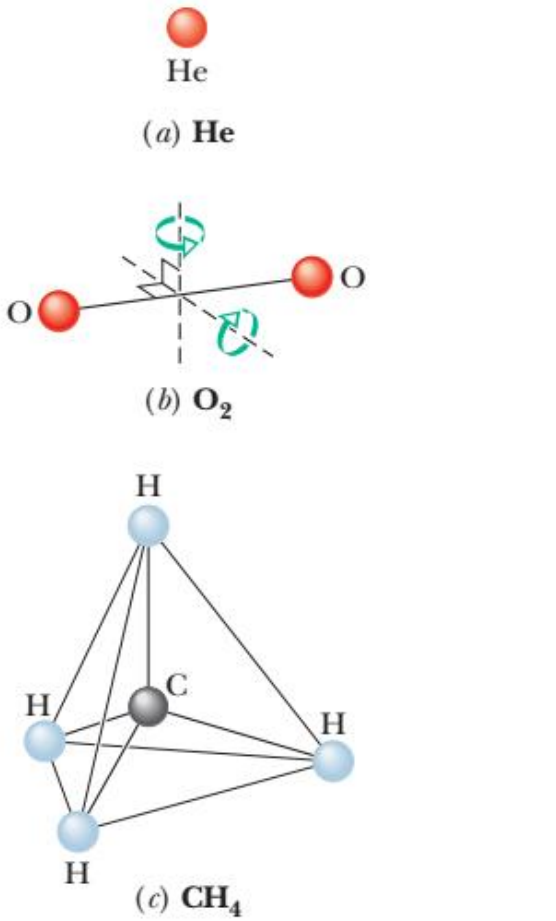
How do we measure specific heat of a substance?

Degrees of freedom and molar specific heats



- Molecules with more than one atom can store E_{int} in forms other than translational kinetic energy.
- Considering He (monatomic gas), O_2 (diatomic gas), and CH_4 (polyatomic gas)
- All three can have translational motions (moving left–right and up–down) and rotational motions (spinning about an axis like a top).
- Diatomic and polyatomic molecules can have oscillatory motions, with the atoms oscillating slightly toward and away from one another.

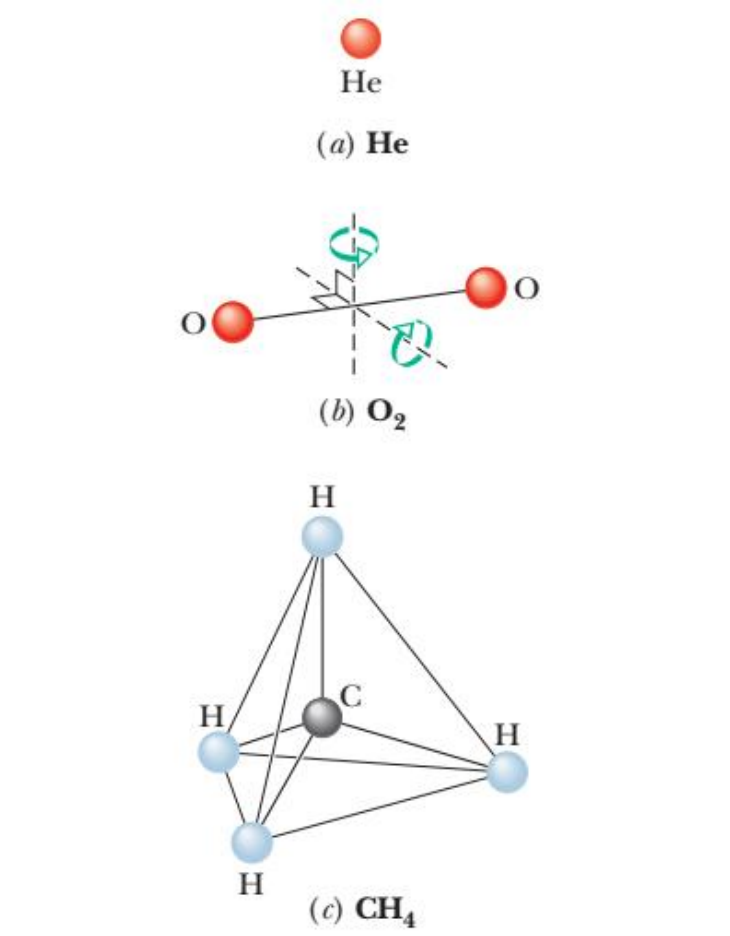
Degrees of freedom and molar specific heats



To keep account of the various ways in which energy can be stored in a gas, James Clerk Maxwell introduced the theorem of the equipartition of energy:

"Every kind of molecule has a certain number f of degrees of freedom, which are independent ways in which the molecule can store energy. Each such degree of freedom has associated with it—on average—an energy of 1/2kT per molecule (or 1/2RT per mole)"

Degrees of freedom and molar specific heats

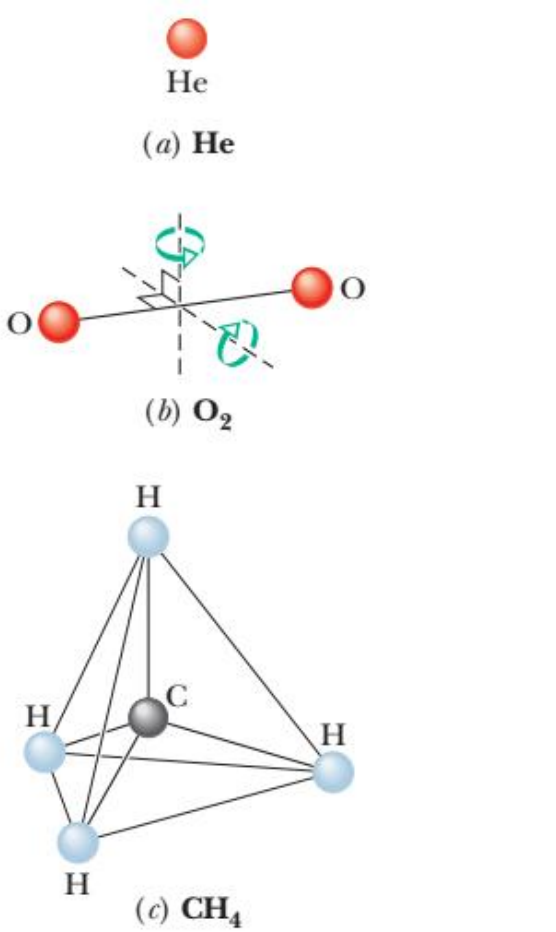


Let's apply the theorem to the motions of the molecules!!

- Monatomic molecules: 3 degrees of translational freedom (3 ways to move in translation along 3 axes → an associated energy of $3(1/2kT)$ per molecule
- Diatomic molecules: 3 degrees of translational freedom + 2 degrees of rotational freedom (can rotate like a top only about axes perpendicular to the line connecting the atoms) → an associated energy of $5(1/2kT)$ per molecule
- *E_{int} can be then calculated as:* $E_{int} = (f/2)nRT$

$$\rightarrow C_v = \left(\frac{f}{2}\right)R = 4.16f\ J/mol.K$$

Degrees of freedom and molar specific heats



Molecule	Example	Degrees of Freedom: Translational	Degrees of Freedom: Rotational	Degrees of Freedom: Total (f)	Predicted Molar Specific Heats: C_V (Eq. 19-51)	Predicted Molar Specific Heats: $C_p = C_V + R$
Monatomic	He	3	0	3	$\frac{3}{2}R$	$\frac{5}{2}R$
Diatomic	O_2	3	2	5	$\frac{5}{2}R$	$\frac{7}{2}R$
Polyatomic	CH_4	3	3	6	$3R$	$4R$

- This prediction agrees with experiment for diatomic gases (f = 5), but it is too low for polyatomic gases
- E.g. CH_4: 3 degrees of translational freedom + 3 degrees of rotational freedom → an associated energy of $6(1/2kT)$ per molecule

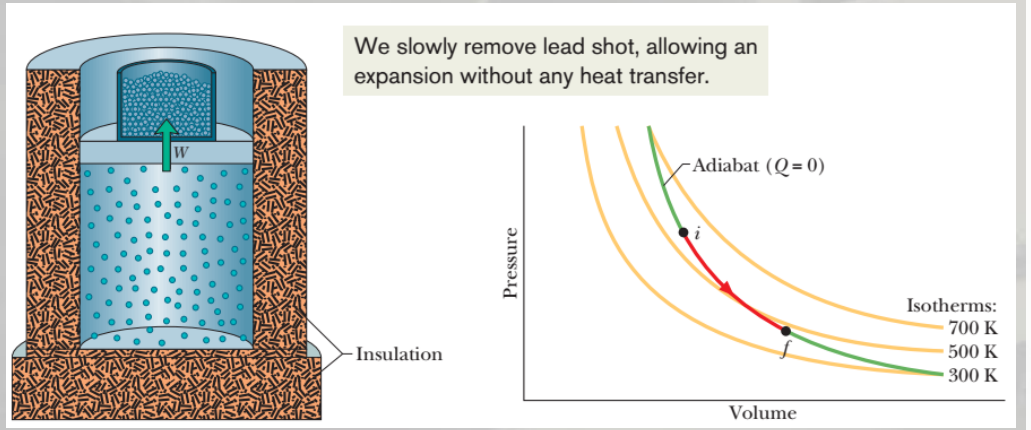
Degrees of freedom and molar specific heats

1. What is the value of ratio between Cp and Cv for a gas of monatomic molecules?

Answer: For a monatomic gas:
$Cv = fR/2 = 3R/2$, $Cp = (2+f)R/2 = 5R/2 \rightarrow Cp/Cv = 5/3$

2. What is the value of molar specific heat of a diatomic gas at constant pressure?
Answer: For a diatomic gas:
$Cv = fR/2 = 5R/2$, $Cp = (2+f)R/2 = 7R/2$

Adiabatic expansion of an ideal gas



- First law for a tiny volume change (P = const):

$$dE_{int} = Q - dW = Q - PdV$$

- Substituting:

$Q = 0$ (no heat transfer)

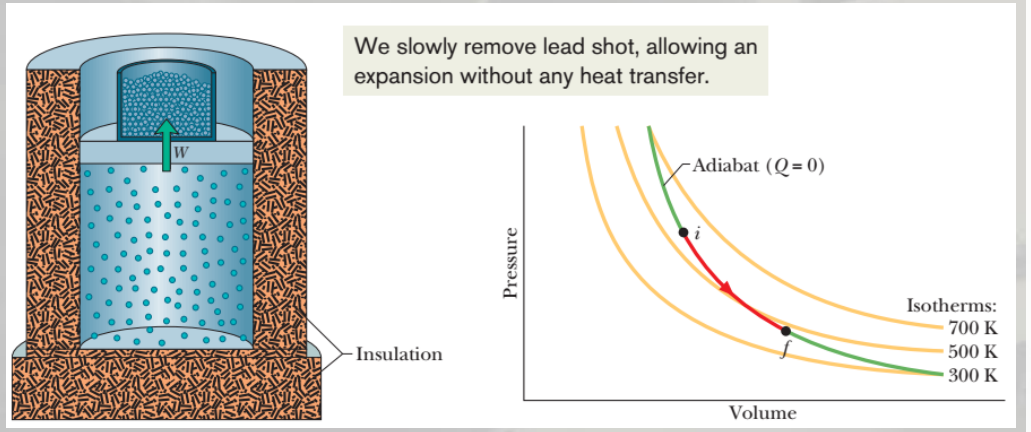
$$dE_{int} = nC_v dT$$

- Then:

$$nC_v dT = 0 - PdV$$

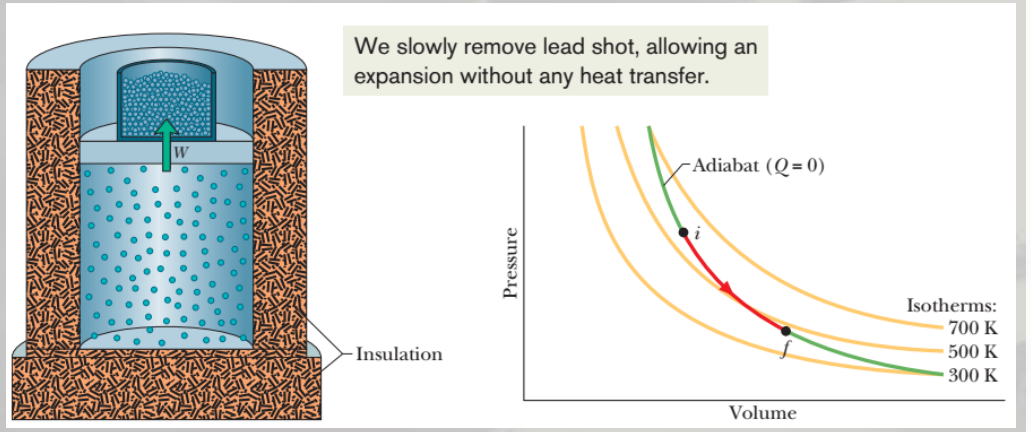
$$ndT = -\frac{P}{C_v} dV$$

Adiabatic expansion of an ideal gas



- From the ideal gas law:
  $d(PV) = d(nRT) \rightarrow VdP + PdV = nRdT$
- Substituting $R = C_p - C_v$:
  $$ndT = \frac{VdP + PdV}{C_p - C_v}$$
- Then:
  $$ndT = -\frac{P}{C_v}dV = \frac{VdP + PdV}{C_p - C_v}$$
  $$-\frac{P}{C_v}dV(C_p - C_v) = VdP + PdV$$
  $$\frac{dP}{P} + \frac{C_p}{C_v}\frac{dV}{V} = 0$$

Adiabatic expansion of an ideal gas



- Subtituting $\gamma = C_p/C_v$ and integrating:

$$lnP + \gamma lnV = a\ const$$

$$lnPV^{\gamma} = a\ const$$

$$e^{lnPV^{\gamma}} = a\ const$$

- Finally: $PV^{\gamma} = a\ const$

$$P_i V_i^{\gamma} = P_f V_f^{\gamma}$$

- Subtituting $P = nRT/V$:

$$\frac{nRTV^{\gamma}}{V} = a\ const \rightarrow TV^{\gamma-1} = a\ const$$

$$T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1}$$

- Remark: For an adiabatic free expansion:

$T_i = T_f$ *(above equations cannot be applied)*

Constant Quantity	Process Type	Some Special Results ($\Delta E_{int} = Q - W$ and $\Delta E_{int} = nC_V \Delta T$ for all paths)
p	Isobaric	$Q = nC_p \Delta T; W = p \Delta V$
T	Isothermal	$Q = W = nRT \ln(V_f/V_i); \Delta E_{int} = 0$
pV^γ, $TV^{\gamma-1}$	Adiabatic	$Q = 0; \quad W = -\Delta E_{int}$
V	Isochoric	$Q = \Delta E_{int} = nC_V \Delta T; \quad W = 0$

Four special processes that an ideal gas can undergo



Exercise

Initially an ideal diatomic gas has pressure P_i = 2.00 x10^5 Pa and volume V_i = 4.00 x 10^{-6}m^3. How much work W does it do, and what is the change in its internal energy if it expands adiabatically to volume V_f = 8.00 x 10^{-6}m^3? Throughout the process, the molecules have rotation but not oscillation.



Exercise

Ideal diatomic gas, $P_i = 2.00 \times 10^5$ Pa, $V_i = 4.00 \times 10^{-6}\text{m}^3$, $V_f = 8.00 \times 10^{-6}\text{m}^3$, $\gamma = \frac{c_p}{c_v} = \frac{\frac{7}{2}R}{\frac{5}{2}R} = 1.4$

a) *W?*

$$PV^{\gamma} = a\ const \rightarrow PV^{\gamma} = P_i V_i^{\gamma}$$

$$P = V^{-\gamma} P_i V_i^{\gamma}$$

$$W = \int_{V_i}^{V_f} P dV = \int_{V_i}^{V_f} V^{-\gamma} P_i V_i^{\gamma} dV$$

$$= P_i V_i^{\gamma} \int_{V_i}^{V_f} V^{-\gamma} dV = \frac{1}{-\gamma + 1} P_i V_i^{\gamma} (V^{-\gamma+1}) \Big|_{V_i}^{V_f}$$

$$= 0.48J$$

b) ΔE_{int}*?*

$$\Delta E_{int} = Q - W = 0 - W = -0.48J$$

Summary

- Concept of kinetic theory of gases
- Ideal gas law ($PV = nRT$)
- Work done by an ideal gas
- Pressure, Temperature, and Molecular Speed
- Temperature and Kinetic Energy
- Mean Free Path
- Maxwell Speed Distribution
- Molar Specific Heats
- Degrees of Freedom and CV
- Adiabatic process

